Instrument Rating Course (IRC) - Instrument Airplane Training Course Outline

This TCO meets the curriculum requirements for the Instrument Rating Course contained in Part 141 Appendix C.

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1 Training Course Outline (TCO)

Record of Revisions

Revision Number	Effective Date	Date Entered	Entered By
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2	05/12/11		
3	11/23/11		
4	03/13/12		
5	07/31/12		
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10	05/17/18		
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12	11/01/21		
13	08/01/23		
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Revision System

Instrument TCO revisions will be issued in consecutive numbers; and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the Instrument TCO to ensure that all current procedures are being followed.

1.1 Aircraft

a. The aircraft that may be used in this course are:

   i. Cessna 172 (4 place)
   ii. Cirrus SR-20 (4 place)
   iii. Piper PA28 (4 place)

b. All aircraft used in this course comply with 14 CFR § 141.39 and must:

   i. Be registered as a civil aircraft of the United States and certified in the standard airworthiness category.
   ii. Be maintained and inspected in accordance with the requirements of 14 CFR Part 91.
   iii. Have engine power controls and flight controls easily accessible and that operate in a normal manner from each pilot station.

1.1.1 Required Aircraft Equipment

a. Transponder with Mode S
b. Equipped with ADS-B IN and OUT or OUT
c. MFD and PFD or standard instrument package
d. GPS or ADF
e. Autopilot (If installed)

1.2 Personnel

1.2.1 Chief Flight Instructor

The Chief Flight Instructor designated for this course is specified in WebOPSS paragraph A006. The Chief Flight Instructor meets the requirements outlined in 14 CFR § 141.35.

1.2.2 Assistant Chief Flight Instructors

The Assistant Chief Flight Instructor(s) designated for this course are specified in WebOPSS paragraph A006.

The Assistant Chief Flight Instructor(s) designated for this course must meet the requirements outlined in 14 CFR § 141.36 and will be designated by letter.

1.2.3 Flight Instructors

Each flight instructor assigned to this course must have completed Flight Instructor Training and be the holder of at least a Commercial Pilot Certificate with an Airplane category and Single Engine Land or Multi Engine Land class rating, as appropriate. In addition, each instructor must hold a Flight Instructor Certificate with an Airplane category and Single Engine class rating or Multi Engine class rating, as appropriate, with an Instrument Rating.

1.2.4 Chief Ground Instructor

The Chief Ground Instructor for this course is specified in WebOPSS paragraph A006. The Chief Ground Instructor meets the requirements outlined in 14 CFR § 141.35(e).

Whenever a Chief or Assistant Chief Ground Instructor is either undesignated or unavailable, the appropriate Chief or Assistant Chief Flight Instructor may assume those duties.

1.2.5 Assistant Chief Ground Instructors

The Assistant Chief Ground Instructor(s) designated for this course are specified in WebOPSS paragraph A006.

The designated Assistant Chief Ground Instructor(s) for this course meets the requirements outlined in 14 CFR § 141.36(e) and will be designated by letter.

1.3 Student Information

1.3.1 Course Enrollment

To be enrolled in this course, a crewmember must hold a current private pilot certificate with at least a third class medical certificate prior to beginning the flight training portion of the course.

1.3.2 Requirements for Graduation

Crewmembers must satisfactorily complete the training outlined in this syllabus and pass all Stage Checks and End of Course Tests.

1.3.3 Lesson Description and Stages of Training

Each lesson is fully described within the syllabus, including the objectives, standards, and measurable units of accomplishment and learning. The stage objectives and standards are described at the beginning of each stage within the syllabus.

1.3.4 Tests and Checks

The syllabus incorporates stage checks and End of Course Tests in accordance with 14 CFR Part 141. These checks are given by the Chief Instructor, a designated Assistant Chief, or another designated Check Instructor, as appropriate.

1.4 Course Overview

1.4.1 Course Objective

The student will obtain the knowledge, skill, and aeronautical experience necessary to meet the requirements for an Instrument Rating Airplane.

1.4.2 Course Completion Standards

The student will demonstrate through written, oral, and flight tests that the knowledge, skill, and experience requirements necessary to obtain an Instrument Rating Airplane are met.

1.4.3 Training Syllabus

The syllabus used will be a combination of the Gleim ground training syllabus and the Acron Aviation Academy Inc. flight training syllabus.

Acron will be utilizing the FAA approved Part 141 Gleim ground training syllabus, seventh edition, for the Instrument Rating Certification Course (Chapter 2 of this TCO). The flight training syllabus will be the most current revision of the Acron Aviation Academy Inc. FAA-approved syllabus (Chapter 3 of this TCO).

The training syllabus meets all curriculum requirements as prescribed by 14 CFR Part 141 and Appendix C. It is composed of separate ground and flight courses to allow greater flexibility and to accommodate individual needs. The training courses may be taught either separately or concurrently.

1.4.4 Training Course

The Ground Training Course contains two (2) stages. The Flight Training Course contains three (3) stages.

> **NOTE:** Students enrolled at Jacksonville University and completing a Bachelor of Science Degree in Aviation Management and Flight Operations receive full credit for all Ground School requirements for the following courses: Private ASEL, Instrument Airplane, Commercial Multi-Engine, CFI-A, and CFI-I.

> **Note:** Students enrolled in the online Gleim Groundschool will be credited for the ground school after successful completion and receipt of the graduation certificate.

1.5 Course Introduction

The Instrument Airplane Course utilizes the building-block theory of learning, which recognizes that each item taught must be presented on the basis of previously learned knowledge and skills.

1.5.1 Course Elements

The Instrument Airplane Course utilizes separate flight and ground segments. The course includes the latest FAA pilot certification requirements and a maximum of student oriented instruction. The syllabus and support materials not only provide necessary information, but also guide the student through the course in a logical manner.

1.5.2 Use of Flight Training Devices

The Instrument Airplane Course allows, as an option, the use of approved flight training devices for those lessons designated in the Course Time Allocation Table under the Aircraft Type heading. The best results are obtained when a flight training device introduction is accomplished just prior to the flight lesson.

1.5.3 Preflight Briefing

The preflight briefing is an integral part of the learning process. During the preflight briefing, the **PUI** (Pilot under training) firstly briefs the **IP** (Instructor Pilot). During this briefing the **PUI** will brief the **IP** on the possible hazards to the route of flight, review current weather, current personal minimums and any other items of importance. Once the **PUI** has conducted their briefing the **IP** must review the items and challenge the information with specific questions. Once this is complete the **IP** will thoroughly brief the **PUI** on the scenario, and all of the listed tasks and completion requirements. The “Preflight Briefing” should take a minimum of 15 minutes per flight/flight training device event.

1.5.4 Postflight Evaluation

The postflight evaluation is equally as important as the preflight briefing. Each scenario must be thoroughly debriefed, during which time the **PUI** will evaluate their performance and debrief the **IP**. The importance of this “Self Evaluation” is to help the **PUI** learn to be critically honest of their own performance and to learn to come up with possible ways to correct any deficiencies without input from the **IP**. The **IP** will then debrief the **PUI** while emphasizing areas where the **IP** saw deficiencies, or where there was a notable difference in the “Self Evaluation” by the **PUI** and the evaluation of the **IP**. Noticeable advancement should be apparent and recommendations should be made for improvement, where appropriate. This action is a valuable instructional technique because it increases retention and helps to prepare the student for the next lesson. The “Postflight Briefing” should take a minimum of 15 minutes per flight/flight training device event.

1.5.5 Student Stage Checks

Stage checks measure the student's accomplishments during each stage of training. The conduct of each stage check is the responsibility of the Chief Instructor. The authority to conduct these tests may be delegated to an appropriately designated and approved Assistant Chief Instructor or Check Instructor. This procedure provides close supervision of training and another opinion on the student's progress. The stage check also gives the Chief Instructor an opportunity to check the effectiveness of the instructors and their teaching methods.

An examination of the building-block theory of learning will show that it is extremely important for progress and proficiency to be satisfactory before the student enters a new stage of training.

Therefore, the next stage should not begin until the student successfully completes the current stage. Failure to follow this progression may defeat the purpose of the stage check and lead to overall course breakdown.

1.5.6 Ground Training

Acron Aviation Academy will be utilizing the FAA-Approved Part 141 Gleim Ground training syllabi for the Instrument Rating Course.

The course may be conducted as part of a formal ground school, or completed using the Gleim FAA-Approved 141 Online Ground School, in compliance with 14 CFR 141, Appendix C, par. 3.

1.5.7 Using the Ground Lessons

Each ground lesson covers the assigned material in the Gleim Instrument Pilot Knowledge Test Prep book, Gleim Pilot Handbook, or online courseware, answering questions in the corresponding study units from the appropriate Gleim FAA Knowledge Test Prep book or courseware, and review of supporting materials as required as referenced in the syllabus. The program is divided into three (2) stages, each requiring a knowledge test before progressing to the next stage. Upon completion of the 3 stages, there is a comprehensive end-of-course knowledge test. After successful completion of the comprehensive knowledge test, the student may be recommended by our staff, or students completing the Gleim Online Ground School will receive a recommendation from the Gleim Chief Instructor for the FAA private pilot knowledge test (IRA).

1.5.8 Textbooks

Prior to each ground lesson, the student should read and study the appropriate textbook section or chapter, as assigned. The appropriate textbook section or chapter will contain the information necessary to help the student complete the academic stages of the Instrument Airplane Course.

1.5.9 DVD Presentations

Following the ground lesson presentations, the student, as an option, may view DVD presentations. After the DVD presentation, the student should feel free to ask questions and obtain clarification of subject material viewed.

1.5.10 Final Examination

When all appropriate ground lesson presentations are complete, the student must take the Training Course Final Exam. Following the test, the instructor should assign the student review of appropriate subject areas. After a thorough review, the actual FAA Knowledge Test, if applicable, should be taken.

Atfer sucessful completion of the Gleim Comprehensive knowledge test, the student may be recommended by the Acron Aviation Chief Flight Instructror or completeting the Gleim Online Ground school will receive a reccomendation from the Gleim Chief Instructor for the FAA Instument knowledge test.

1.5.11 Instrument Airplane Course Implementation

The lesson sequence and content have been designed to provide the student with maximum training prior to the introduction of new maneuvers or procedures.

If any deviation from the normal lesson progression is desired, only the Chief Instructor or designated Assistant Chief or Flight Training Manager for the course may grant approval.

> **NOTE:** If the student is completing the ground training and flight training courses concurrently, the instructor must ensure that the student's flight training does not progress faster than the ground training.

1.6 Sanford Facility (SFB)

Acron Aviation Academy's Sanford base is located at the Orlando Sanford International Airport, 2700 Flightline Avenue, Sanford, FL 32773.

1.6.1 Chief and Designated Assistant Chief Flight Instructor

The Chief Flight Instructor for Acron Aviation Academy is designatged in WebOPSS paragraph A006.

1.6.2 Airport

The Orlando Sanford International Airport (SFB) is the main operations base for training. The airport has four (4) hard surface runways and meets the requirements of 14 CFR Part 141.38 for day and night flight operations.

A federally operated control tower operates daily between 0630-2300 hours. The airport has ILS, and GPS approaches. Fuel and maintenance services are available daily between 0600-2300 hours.

1.6.3 Training Airports

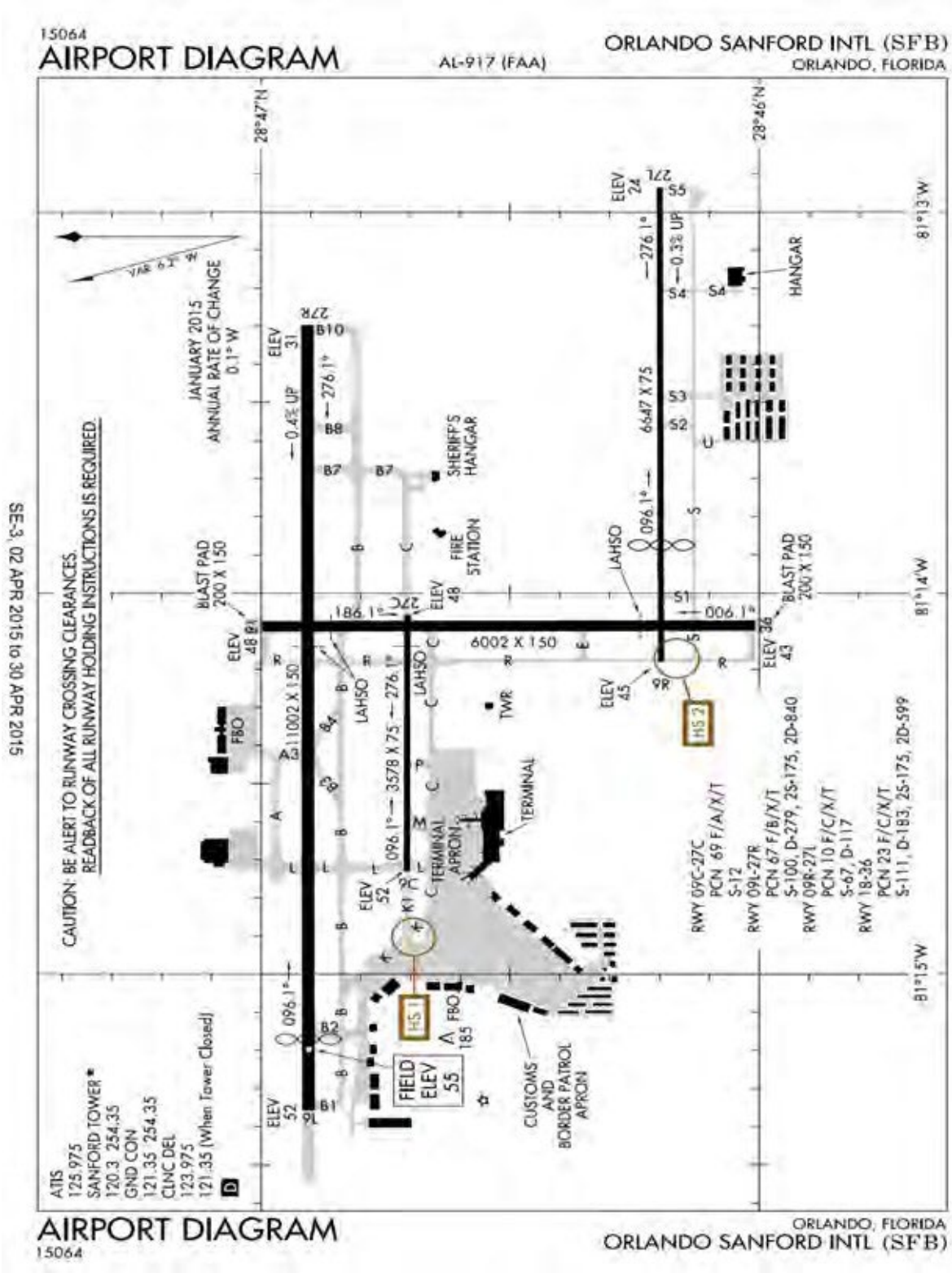
All training airports meet the requirements outlined in 14 CFR § 141.38.

Direction is given to the Academy's flight instructors through an Approved Airports list located in the company operations manual.

The Chief Flight Instructor or designated Assistant Chief Flight Instructor may approve any appropriate public use airport listed in the Chart Supplement.

1.6.4 Sanford Airport Diagram

*NOTE*Airport diagram is not current and not to be used of navigation.



1.6.5 Operations Center

Acron Aviation Academy's Operations Center is the focal point for all flight lesson activity. Adequate areas have been provided for preflight briefings, dispatch, and postflight briefings of all flight lesson activities.

The Operations Center includes a Planning Area and Briefing Rooms. The Flight/Ground Briefing Center provides additional space for briefings and small group instruction. Each briefing area is equipped with an erasable marker board as well as tables and chairs to accommodate students and instructors.

1.6.6 Planning Area

This area is equipped with tables and chairs adequate for both students and instructors.

Telephones are available to provide communication to the nearest FAA Flight Service Station. In addition, weather stations are also available for preflight planning purposes.

1.6.7 Sanford Simulator Training Building

Acron Aviation Academy's Simulator Training Building is the focal point for all simulator and flight training device activity. Adequate areas have been provided for AATD and FTD training lessons.

The Simulator Training Building includes Briefing Rooms. Each briefing area is equipped with an erasable marker board as well as tables and chairs to accommodate students and instructors.

> **NOTE:** All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

1.6.8 Academics

The Academics facilities are located in the Academics Building. This building houses classrooms, and the Learning Resource Center.

1.6.9 Classrooms

Classrooms are equipped with tables and chairs to accommodate students. Each classroom contains at least one (1) chalkboard or erasable marker board of adequate size and one (1) projection screen.

1.6.10 Learning Resource Center

The Learning Center (LRC) is equipped with tables and chairs to accommodate students. The LRC is available to students wishing to accomplish additional studying in a group or with a tutor. In addition, the LRC provides computer stations to students wishing to accomplish study using CBT materials, or by the viewing of DVD's.

1.6.11 Computer Testing Building

The Computer Testing Building is located at Bldg 41 2749 Flightline Ave, Sanford, FL 32773. The Computer Testing Building has been set aside for the purpose of administering computerized FAA Knowledge Test Examinations.

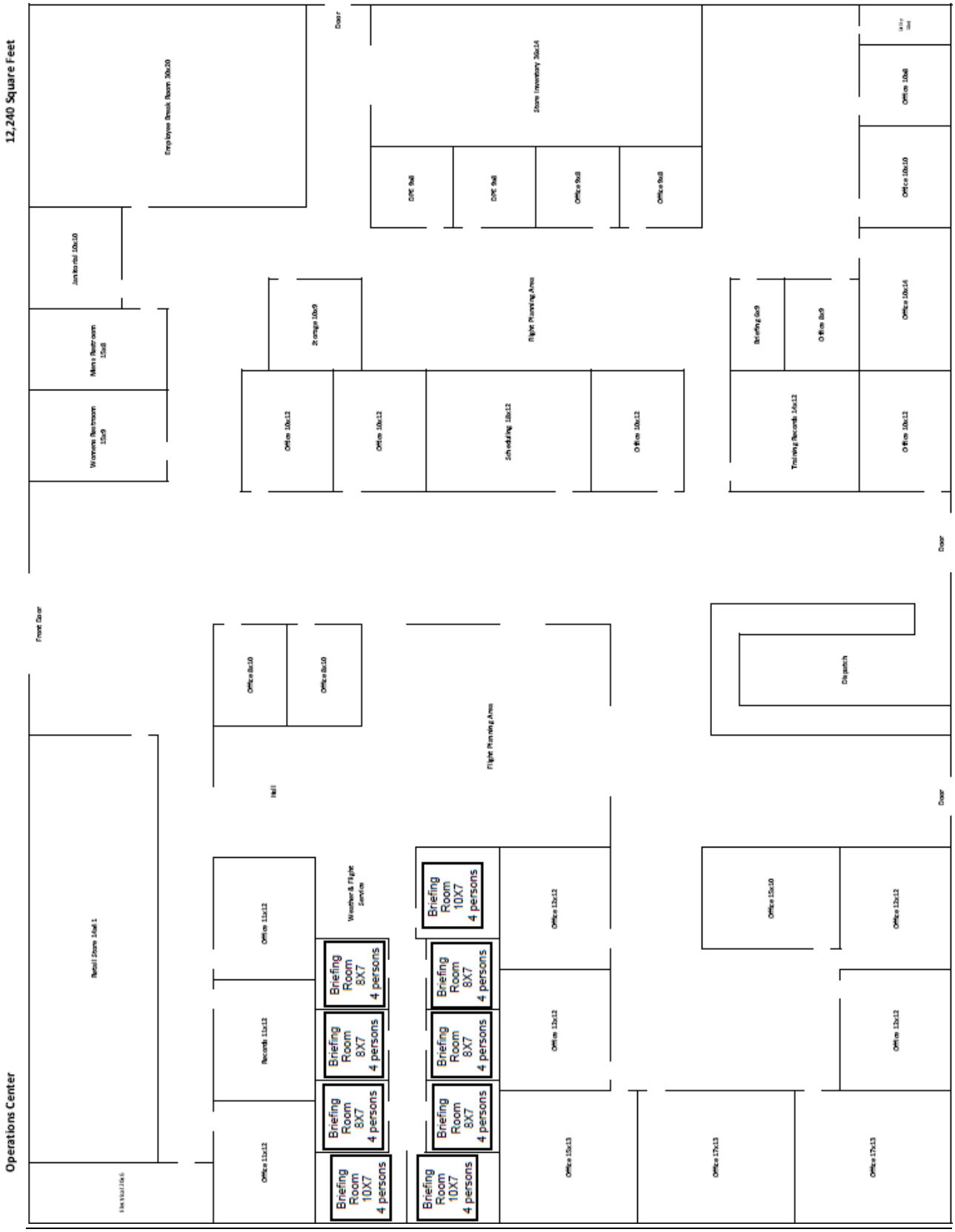
> **NOTE:** All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

Sanford Operations Building –

East section now included on following page in complete building schematic

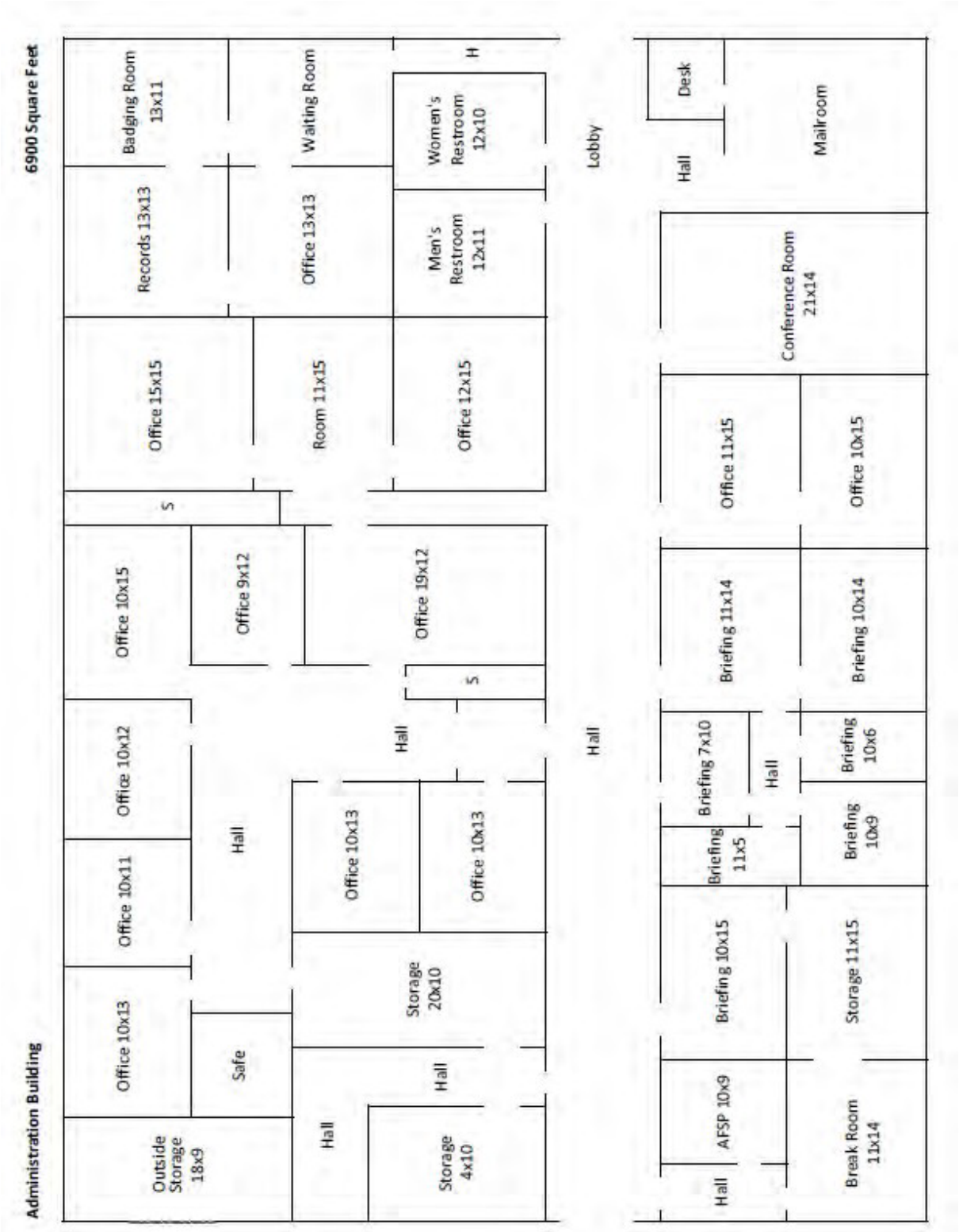
1.6.12 Building Diagrams

1.6.12.1 Sanford Operations Building –

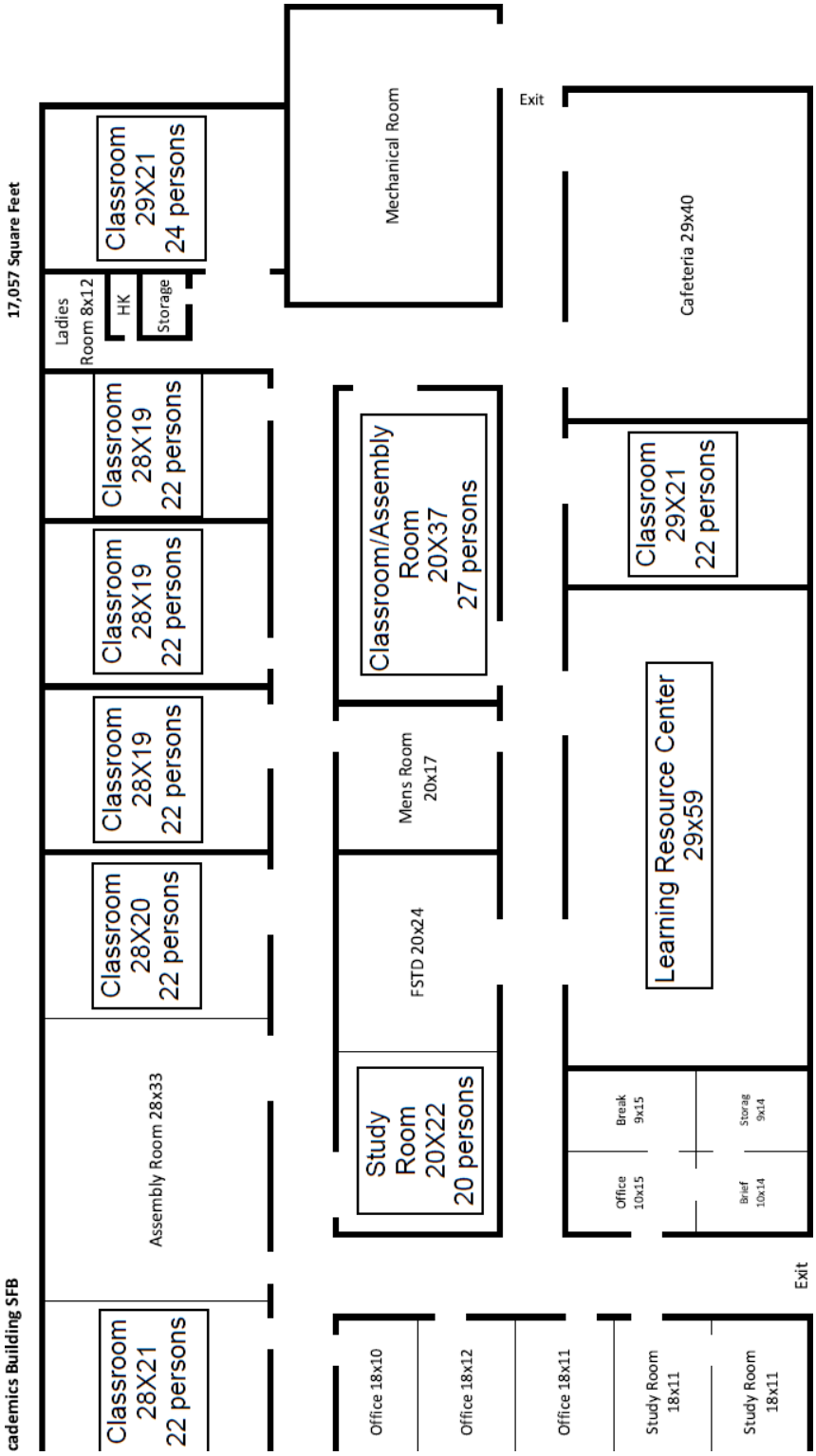


1.6.12.2 Sanford Administration Building

NOTE Building is currently not in use for academy operations.



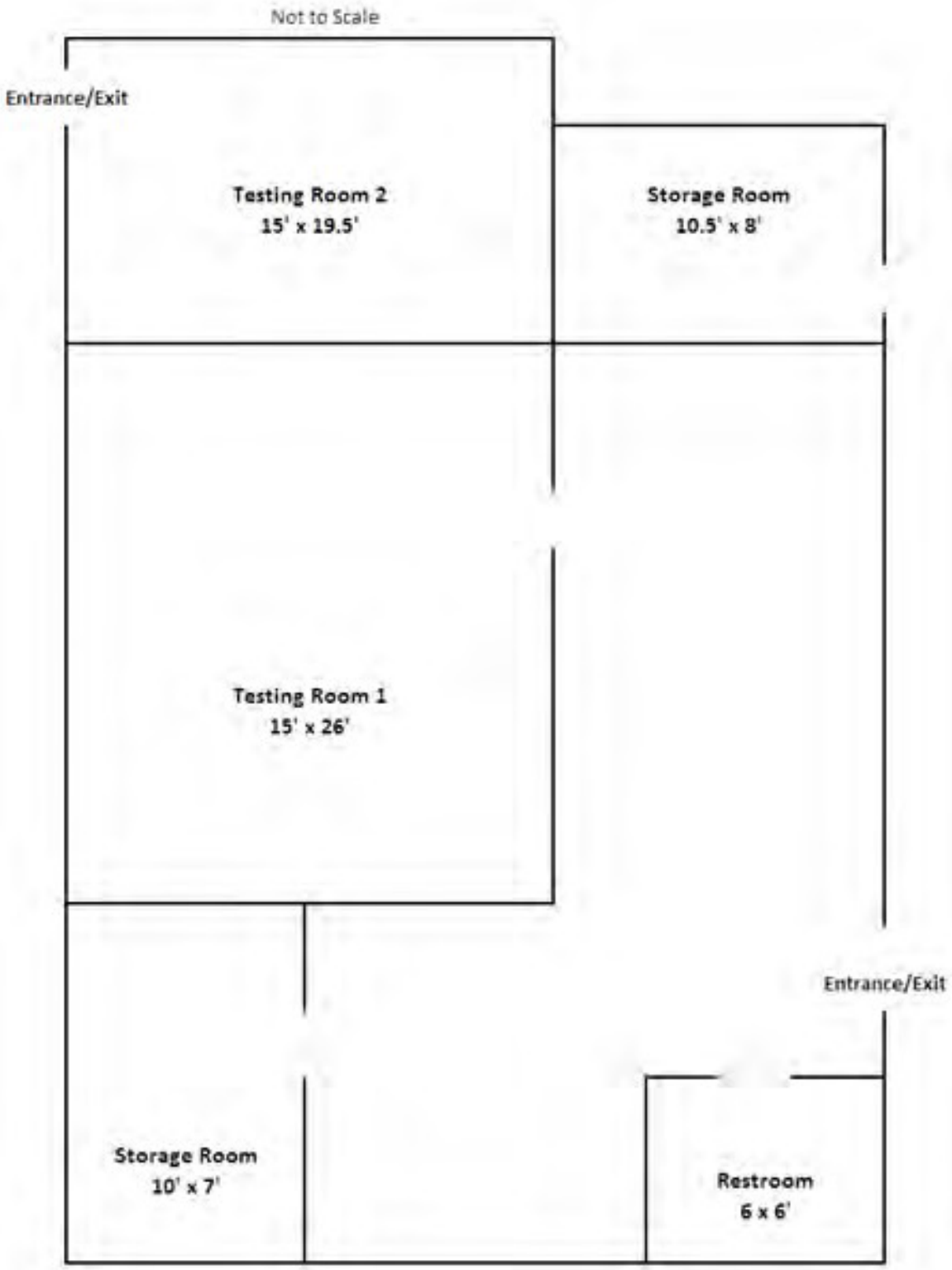
1.6.12.3 Sanford Academics Building



1.6.12.4 Sanford Simulator Training Building

1.6.12.5 Sanford Computer Testing Building

NOTE Testing room 1 has 10 stations for student testing Per PSI standards.



1.6.13 Equipment

1.6.13.1 Ground Training Aids

The training aids available for use, as appropriate, are as follows:

a. Audiovisual Aids

- Computers with projectors, overhead projectors, Interactive Smart Boards, Diagrams and Charts

b. Mockups

- Large, presentation-type, E6B flight computer and Cockpit Procedures Trainer

c. Aircraft Components

- Flight instruments and aircraft engines on stands

In addition, for enhancement and reinforcement of training a variety of video tapes and computer-based training is available.

1.6.13.2 Flight Training Devices

The Academy's flight training devices/approved aviation training devices meet the requirements specified in 14 CFR § 141.41 (b)/14 CFR § 141.41 (c) respectively, and 14 CFR Part 60. These devices may be used to meet the total simulated flight time allowed in 14 CFR Part 141, Appendices B through G and Appendix I.

a. The Aerosim-Mechtronix Level 6 flight training device (FTD) is an enclosed stationary trainer equipped with an exact replica of the aircraft that meets FAA Level 6 certification criteria. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR § 91.205 (b) and (d) is provided or simulated.
b. Aerosim-Mechtronix Advanced Aviation Training Device (AATD). The Aerosim Model Aviator 100 Single Engine Land is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR § 91.205 (b) and (d) is provided or simulated.
c. The Frasca International Inc. TruFlite Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR § 91.205 (b) and (d) is provided or simulated.
d. The Redbird MCX Advanced Aviation Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically advanced, single and multi-engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR § 91.205 (b) and (d) is provided or simulated.
e. The Frasca International Inc. RTD Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR § 91.205 (b) and (d) is provided or simulated.

1.7 Jacksonville Facility (CRG)

Acron Aviation Academy's Jacksonville satellite base is located at the Jacksonville Executive At Craig Airport, 793 Amelia Earhart Drive, Jacksonville, FL 32225.

1.7.1 Chief and Designated Assistant Chief Flight Instructor

The Chief Flight Instructor for Acron Aviation Academy is designated in WebOPSS paragraph A006. The designated Assistant Chief Flight Instructor for the Jacksonville Base is specified in WebOPSS paragraphs A006 and B050.

1.7.2 Airport

The Jacksonville Executive At Craig Airport (CRG) is the satellite base's main location for training. The airport has two (2) hard surface runways and meets the requirements of 14 CFR Part 141.38 for day and night flight operations.

An FAA contract control tower operates between 0600-2300 Monday through Friday and 0700-2200 Saturday and Sunday.

The airport has ILS, VOR and GPS approaches.

Fuel service is available from 0600-2200 hours daily. Maintenance service is available full time from 600-1900 (M-F), 0900-1700 (Saturday) and on call other times.

1.7.3 Training Airports

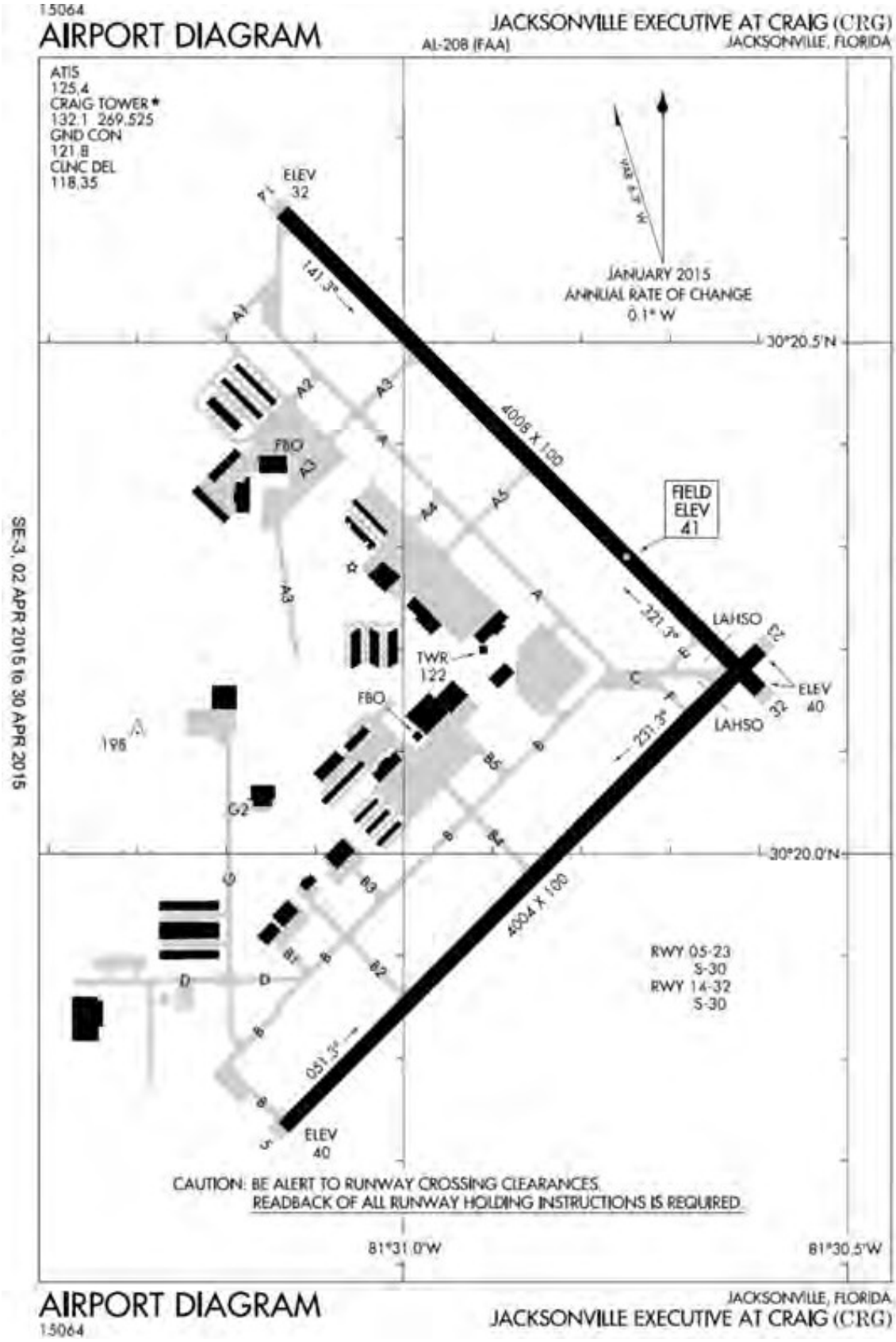
All training airports meet the requirements outlined in 14 CFR § 141.38.

Direction is given to the Academy's flight instructors through an Approved Airports list in supplemental documentation.

The Chief Flight Instructor or designated Assistant Chief Flight Instructor may approve any appropriate public use airport listed in the Airport/Facilities Directory.

1.7.4 Jacksonville/Craig Airport Diagram

NOTE Diagram not current and not to be used for navigational purposes.



1.7.5 Operations Center

Acron Aviation Academy's Operations Center is the focal point for all flight lesson activity. Adequate areas have been provided for the dispatch and recovery of all flight lesson activity.

The Operations Center has adequate Pre-Post Flight Briefing Areas.

1.7.6 Pre/Post Flight Briefing Areas

Briefing Rooms 1 and 2 are equipped with tables and chairs adequate for both students and instructors. Briefing Room 1 may accommodate up to 16 students. Briefing Room 2 may accommodate up to 8 students. Briefing Room 2 may be reserved for FAA practical test oral examinations and stage checks as needed. All briefing areas are also equipped with erasable marker boards.

A telephone is available to provide communication to the nearest FAA Flight Service Station.

1.7.7 Simulator Hall

The Jacksonville facility has s Simulator Hall. T h e Sim Hall 1 includes AATD simulators.

1.7.8 Equipment

1.7.8.1 Ground Training Aids

The training aids available for use, as appropriate, are as follows:

a. Audiovisual Aids

- Computers, Diagrams and Charts

b. Mockups

- Cockpit Procedures Trainer

c. Aircraft Components Flight instruments

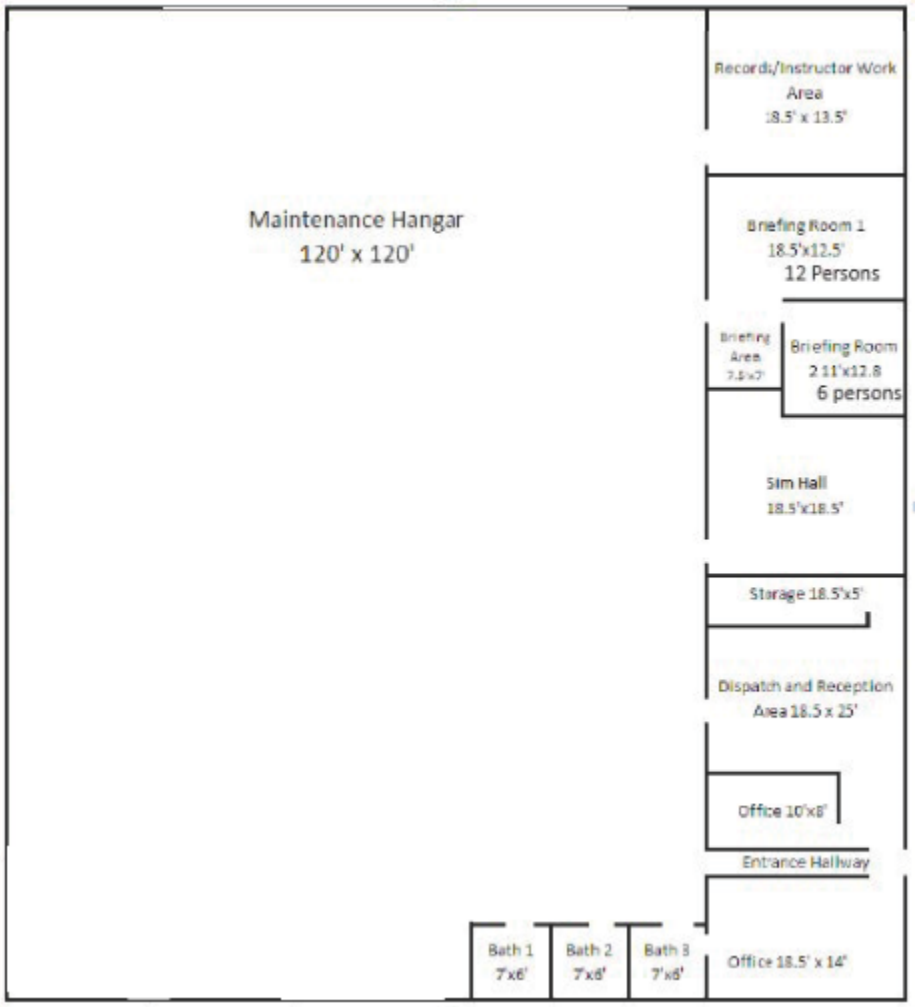
In addition, for enhancement and reinforcement of training a variety of video tapes and computer-based training is available.

1.7.8.2 Flight Training Devices

The Academy's flight training devices/approved aviation training devices meet the requirements specified in 14 CFR § 141.41 (b)/14 CFR § 141.41 (c) respectively, and may be used to meet the total simulated flight time allowed in 14 CFR Part 141, Appendices B through G and Appendix I.

The Frasca International Inc. RTD Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane or multi engine (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.

1.7.8.3 Operations Center Diagram



1.7.8.4 Academics

The Academics facilities are located on the campus of Jacksonville University at 2800 University Blvd. N., Jacksonville, FL 32211.

1.7.8.5 Classrooms

Classroom D-153 in the Davis College of Business Building may accommodate 41 students. Depending on the size of the class and the scheduling needs of the college other rooms may be assigned.

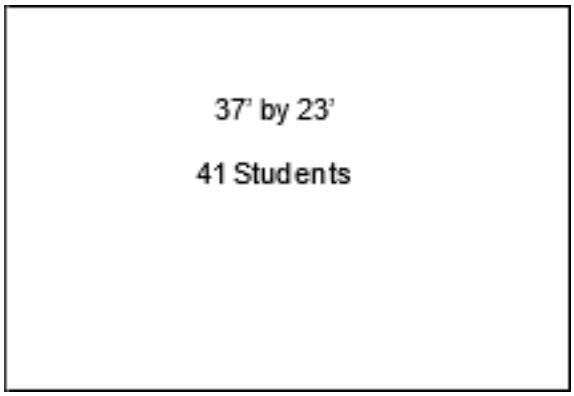
1.7.8.6 Ground Training Aids

a. PC/DVD/Internet with overhead projection capability
b. Computer-based training
c. Flight instruments
d. Static displays of aircraft engines and parts Aircraft models
e. GiantE6B White board

NOTE: All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

1.7.8.7 Jacksonville University Classroom Diagram

Davis College of Business Room D-153



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2 Instrument Airplane Ground Syllabus

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Chapter 2					
2-5	8	04/07/2025			
2-6	8	04/07/2025			

FAA APPROVED

PAULINE AYN O'CONNOR

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ORLANDO FSDO-SO15

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2 Instrument Airplane Ground Syllabus

Record of Revisions

Revision Number	Effective Date	Date Entered	Entered By
Issue	09/09/07		
1	02/07/08		
1.1	03/02/09		
2	09/03/09		
3	12/08/10		
4	03/13/12		
5	05/17/17		
6	09/07/20		
7	08/01/23		
8	04/07/2025		
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Revision System

Instrument Airplane Course revisions will be issued in consecutive numbers and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the Instrument Airplane Course to ensure that all current procedures are being followed.

2.1 Instrument Airplane Course Objective

The objective of this course is to introduce the Pilot Under Instruction (PUI) to the Instrument Airplane aviation-training environment. The PUI will study all aeronautical knowledge areas required for an Instrument Airplane Rating. This course is designed for an applicant that already holds a Private Pilot Airplane Single Engine Land Certificate.

a. The Instructor will conduct training using the following equipment:

   i. Classroom
   ii. Computer
   iii. Projector
   iv. Whiteboard

b. Reference Textbooks for this course include:

   i. Gleim Instrument Pilot FAA Knowledge Test Prep
   ii. Gleim Instrument Pilot Flight Maneuvers and Practical Test Prep
   iii. Gleim Instrument Pilot ACS and Oral Exam Guide
   iv. Gleim Aviaiton Weather and Weather Services
   v. Gleim Pilot Handbook
   vi. Gleim FAR/AIM
   vii. Online access to Gleim Instrument Rating

2.2 Instrument Airplane Course Completion Standards

At the completion of this course, the PUI will demonstrate to the Instructor a basic understanding of all aeronautical knowledge areas required for an Instrument Airplane Rating. The PUI will be able to accomplish the objectives of the course to the appropriate level of performance, as outlined in the examination completion standards.

Upon successful completion of the ground training syllabus, the student must complete the FAA knowledge exam for the Instrument Rating. Students who have completed the exam prior to enrollment will not have to retake the exam, provided the test report is still valid.

> **NOTE:** All references in the Gleim syllabus pertaining to the Gleim flight syllabus are not applicable. The Acron Aviation training syllabus (Chapter 3) is the syllabus used.

2.3 Gleim Ground Syllabus Instrument Rating

The Gleim Ground Syllabus for the Instrument Rating can be found in the Gleim Instrument Knowledge Test prep book or in the on-line course ware (login will be provided by the academy)

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INSTRUMENT AIRPLANE FLIGHT SYLLABUS

3 Instrument Airplane Flight Syllabus

Record of Revisions

Revision Number	Effective Date	Date Entered	Entered By
1	12/08/10		MM/JM
2	03/13/12		SP
3	08/03/12		TM
4	04/02/13		SP
5	05/22/13		SP
6	11/04/15		JL
7	05/17/17		NS
8	09/02/20		NS
9	05/25/21		CG
10	08/01/23		CG
11	04/07/2025		DK
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Revision System

Instrument Rating Syllabus course revisions will be issued in consecutive numbers and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the Instrument Rating Syllabus to ensure that all current procedures are being followed.

3.1 Suggested Course Time Allocation Table

Stage	Lesson	Ground Brief	Pre Brief	Post Brief	AA D/FTD	Aircraft	Dual	X-C	Instrument Time	Stage Check Oral	Stage Check Flight	Lesson Equipment
1	1-GB	1.0										Ground
1	2-GB	1.5										Ground
1	3-AA TD/FTD		0.25	0.3	1.2		1.2		0.8			AA TD/FTD
1	4-AA TD/FTD		0.25	0.3	1.2		1.2		0.8			AA TD/FTD
1	5-DL		0.25	0.5		1.2	1.2		1.5			S.E. Airplane
1	6-AA TD/FTD		0.25	0.3	1.2		1.2		0.8			AA TD/FTD
1	7-DL		0.25	0.3	1.0		1.0		0.8			S.E. Airplane
1	8-DL		0.25	0.5		1.5	1.5		1.5			S.E. Airplane
1	9-AA TD/FTD		0.25	0.3	1.2		1.2		0.8			AA TD/FTD
1	10-AA TD/FTD		0.25	0.3	1.0		1.0		0.8			AA TD/FTD
1	11-DL		0.25	0.5		2.0	2.0		1.7			S.E. Airplane
1	12-GB	1.5										Ground
1	13-AA TD/FTD		0.25	0.3	1.2		1.2		1.0			AA TD/FTD
1	14-AA TD/FTD		0.25	0.3	1.2		1.2		1.0			AA TD/FTD
1	15-AA TD/FTD		0.25	0.3	1.5		1.5		1.0			AA TD/FTD
1	16-DL		0.25	0.5		1.7	1.7		1.6			S.E. Airplane
1	17-AA TD/FTD		0.25	0.3	1.5		1.5		1.0			AA TD/FTD
1	18-STO	2.0								2.0		Ground
1	19-STF		0.25	0.3	1.8		1.8		1.2		1.8	AA TD/FTD
2	20-AA TD/FTD		0.25	0.3	1.0		1.0					AA TD/FTD
2	21-GB	1.5										Ground
2	22-AA TD/FTD		0.25	0.3	1.2		1.2		1.0			AA TD/FTD
2	23-GB	2.0										Ground
2	24-AA TD/FTD		0.25	0.3	1.8		1.8		1.0			AA TD/FTD
2	25-DL		0.25	0.5		2.2	2.2		2.2			S.E. Airplane
2	26-AA TD/FTD		0.25	0.3	1.2		1.2		0.8			AA TD/FTD
2	27-AA TD/FTD		0.25	0.3	1.8		1.8		1.0			AA TD/FTD
2	28-DL		0.25	0.3	1.5		1.5		1.0			S.E.Airplane
2	29-DL		0.25	0.5		2.0	2.0		2.2			S.E. Airplane
2	30-DL		0.25	0.3	1.5		1.5		1.5			S.E.Airlane
2	31-DL		0.25	0.5		2.0	2.0		2.2			S.E. Airplane
2	32-AA TD/FTD		0.25	0.3	1.5		1.5		1.0			AA TD/FTD
2	33-GB	1.5										Ground
2	34-DL		0.25	0.5		2.0	2.0		2.2			S.E. Airplane
2	35-STO	2.0								2.0		Ground
2	36-STF		0.25	0.5		2.5	2.5		2.5		2.5	S.E. Airplane
3	37-GB	1.5										Ground
3	38-DXC		0.25	0.5		2.5	2.5	2.5	2.5			S.E. Airplane
3	39-DXC		0.25	0.5		4.0	4.0	4.0	3.8			S.E. Airplane
3	40-GB	1.5										Ground
3	41-DL		0.25	0.5		2.2	2.2		2.2			S.E. Airplane
3	42- STO	2.5								2.5		Ground
3	43- STF (EOC)		0.25	0.5		2.2	2.2		2.2		2.5	S.E. Airplane
	Totals	**18.5**	**8.0**	**12.2**	**26.1**	**26.7**	**52.8**	**6.5**	**44.0**	**6.5**	**6.8**	

3.1.1 Course Minimums

MINIMUM BRIEFING TIME	INSTRUMENT TRAINING	AIRCRAFT INSTRUMENT
7.5	35	21

NOTE: The individual times shown on this table are for instructor/student guidance only. They are not mandatory for each flight.

Instrument Training Totals from Course Table:

AATD/FTD Instrument: 14.0

Aircraft Instrument: 30.0

Total Course Inst.: 44.0

NOTE: Per FAR 141 training requirements, use of an AATD or FTD cannot exceed 40% of the total requirements for the course.

3.2 Stage 1

3.2.1 Stage Objective

The objective of this stage is for the student to demonstrate precise airplane attitude control solely by instrument reference.

Situations requiring the student to execute emergency/abnormal procedures may be presented by the flight instructor and any time during the lesson.

3.2.2 Stage Completion Standards

At the completion of this stage, the student will demonstrate to the flight instructor an increased level of proficiency in precise airplane attitude control by instrument reference only. This will include the use of full panel and partial panel instrument reference.

> **NOTE**: In the event an AATD/FTD is not available, the training event prescribed can be conducted in the aircraft with the approval from the Chief Instructor/ Asst. Chief or designee.

3.2.3 Lesson 1 - GB

3.2.3.1 Equipment

Briefing Room with Cockpit Poster

3.2.3.2 Lesson Objective

This lesson provides an introduction to instrument flying using basic attitude instrument flying procedures.

3.2.3.3 Lesson Content

Publications	
Instrument Rating ACS	Introduction
Instrument Flying Handbook	Introduction
Aircraft Operating Handbook	Introduction
Flight Standardizations Manual	Introduction
Pilot Qualifications	Introduction
Flight by Reference to Instruments	
Primary and Supporting Method	Introduction
Control/Performance Method	Introduction
Preflight Procedures	
Instrument Flight Deck Check	Introduction
Chair Flying	
Cockpit Flows	Introduction
Instrument Flying Procedures	Introduction
Additional Areas	
As Assigned by Instructor (If Necessary)	

3.2.3.4 Completion Standards

At the completion of this lesson the student will have gained an increased level of knowledge in the tasks listed above.

3.2.4 Lesson 2 – GB

3.2.4.1 Equipment

Briefing Room with Cockpit Poster

3.2.4.2 Lesson Objective

This ground briefing will build on the skills introduced during lesson 1. Ample time should be spent to prepare the student for all future flight training. The flight instructor will also conduct a Preflight Assessment of the aircraft with the student which will to familiarize the student with the aircraft.

3.2.4.3 Lesson Content

Aircraft Performance Data	
Performance Charts	Introduction
V Speeds	Introduction
Takeoff Data Card	
Pressure Altitude	Introduction
Density Altitude	Introduction
Fuel Requirements	Introduction
Takeoff & Landing Distances	Introduction
Operation of Systems	
Engine	
Fuel System	Introduction
Electrical System	Introduction
Primary Flight Controls and Trim	Introduction
Flaps	Introduction
Propeller	Introduction
Landing Gear	Introduction
Hydraulic	Introduction
Avionics	Introduction
Pitot-Static System	Introduction
Vacuum System	Introduction
Environmental	Introduction
Deicing and Anti-Icing	Introduction
Chair flying	
Use/Location of Equipment	Introduction
Cockpit Flows	Introduction
Additional Areas	
As Assigned by Instructor (If Necessary)	

3.2.4.4 Completion Standards

At the completion of this lesson the student will have a solid understanding of the tasks required during the preflight preparation and assessment.

3.2.5 Lesson 3 – AATD/FTD

3.2.5.1 Equipment

AATD/FTD

3.2.5.2 Lesson Objective

This lesson provides an introduction to attitude instrument flying. Proper instrument scan, interpretation, pitch/power coordination and trim technique will be emphasized.

3.2.5.3 Lesson Content

Avionics MFD Setup and Usage (If Installed) PFD Setup and Usage (If Installed) GPS Setup and Usage (If Installed) Flight Director Usage (If Installed)	 Introduction Introduction Introduction Introduction
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb	 Review
Climbs (IR) Constant Airspeed Climbs Constant Rate Climbs	 Introduction Introduction
Flight by Reference to Instruments Instrument Scan (Cross-check) Instrument Interpretation Aircraft Control Straight and Level Flight Change of Airspeed Use of Trim	 Introduction Introduction Introduction Introduction Introduction Introduction
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control	 Introduction Introduction
Turns (IR) Standard Rate Turns Constant Bank Turns Climbing & Descending Turns	 Introduction Introduction Introduction
Descents (IR) Constant Airspeed Descents Constant Rate Descents	 Introduction Introduction
Additional Areas As Assigned By Instructor (If Necessary)	

3.2.5.4 Completion Standards

The student should demonstrate an understanding of the instrument indications related to aircraft control. During this flight the student should maintain altitude within ±200 feet, heading within ±20o, airspeed ±15 knots.

3.2.6 Lesson 4 – AATD/FTD

3.2.6.1 Equipment

AATD/FTD

3.2.6.2 Lesson Objective

This lesson provides a review of attitude instrument flying with special emphasis on learning precise aircraft control solely by instrument reference. Proper instrument scan, interpretation, pitch/power coordination and trim technique will be emphasized.

3.2.6.3 Lesson Content

Preflight Procedures	
Aircraft Systems Related to IFR Operations	Introduction
Aircraft Flight Instruments and Navigation Equipment	Introduction
Instrument Flight Deck Check	Introduction
Preflight Assessment	Introduction
Avionics	
MFD Setup and Usage (If Installed)	Review
PFD Setup and Usage (If Installed)	Review
GPS Setup and Usage (If Installed)	Review
Flight Director Usage (If Installed)	Review
Climbs (IR)	
Constant Airspeed Climbs	Review
Constant Rate Climbs	Review
Flight by Reference to Instruments	
Instrument Scan (Cross-check)	Review
Instrument Interpretation	Review
Aircraft Control	Review
Change of Airspeed Use of Trim	Review
Pattern Alpha	Introduction
Slow Flight (IR)	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Turns (IR)	
Standard Rate Turns	Review
Constant Bank Turns	Review
Climbing &Descending Turns	Review
Descents (IR)	
Constant Airspeed Descents	Review
Constant Rate Descents	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

3.2.6.4 Completion Standards

At the completion of this lesson, the student will demonstrate an understanding of the instrument indications related to aircraft control. During this flight, the student should maintain altitude within ±200 feet, heading within ± 20º, airspeed within ± 15 knots.

3.2.7 Lesson 5 – DL

3.2.7.1 Equipment

S.E. AIRPLANE

3.2.7.2 Lesson Objective

This lesson provides a review of attitude instrument flying with special emphasis on precise aircraft control solely by instrument reference. Proper instrument scan, interpretation, pitch/power coordination and trim technique will be emphasized.

3.2.7.3 Lesson Content

Preflight Procedures	
Aircraft Systems Related to IFR Operations	Review
Aircraft Flight Instruments and Navigation Equipment	Review
Instrument Flight Deck Check	Review
Preflight Assessment	Review
Takeoffs, Landings, and Go-Arounds	
Normal Takeoff and Climb	Review
Normal Approach and Landing	Review
Climbs (IR)	
Constant Airspeed Climbs	Review
Constant Rate Climbs	Review
Flight by Reference to Instruments	
Instrument Scan (Cross-check)	Review
Instrument Interpretation	Review
Aircraft Control	Review
Change of Airspeed	Review
Use of Trim	Review
Pattern Bravo	Introduction
Slow Flight (IR)	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Turns (IR)	
Standard Rate Turns	Review
Constant Bank Turns	Review
Descending Turns	Review
Climbing Turns	Review
Descents (IR)	
Constant Airspeed Descents	Review
Constant Rate Descents	Review
Postflight Procedures	
After Landing, Parking and Securing	Review
Checking Instruments and Equipment	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

3.2.7.4 Completion Standards

At the completion of this lesson, the student will demonstrate an understanding of the instrument references related to aircraft control. The student should maintain altitude within ±200 feet, heading within ± 20º. Climb and descent airspeeds will be maintained within ±15 knots. All Normal Takeoff and Climbs and landings must be safely executed with minimal instructor assistance.

3.2.8 Lesson 6 – AATD/FTD

3.2.8.1 Equipment

AATD/FTD

3.2.8.2 Lesson Objective

This lesson provides a review of attitude instrument flying.

3.2.8.3 Lesson Content

Preflight Procedures Instrument Flight Deck Check	Review
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb Procedures	Review
Flight by Reference to Instruments Straight and Level Flight Change of Airspeed Operations in Turbulence Recovery from Unusual Flight Attitudes	 Review Review Introduction Introduction
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control	 Review Review
Steep Turns (IR) Steep Turns	Introduction
Climbs (IR) Constant Airspeed Climbs Constant Rate Climbs	 Review Review
Turns (IR) Standard Rate Turns Timed Turns to Magnetic Compass/ Standby Instrument Headings Compass Turns to Magnetic Compass Headings (If Installed	 Review Introduction Introduction
Descents (IR) Constant Airspeed Descents Constant Rate Descents	 Review Review
Stalls (IR) Power-Off Stalls Power-On Stalls	 Introduction Introduction
Additional Areas As Assigned By Instructor (If Necessary	

3.2.8.4 Completion Standards

At the completion of this lesson, the student will be able to immediately recognize the approach of stalls and other unusual attitudes and demonstrate correct recovery procedures. Altitude should be maintained within ±150 feet, heading within ±15 o and airspeed within ±15 knots.

3.2.9 Lesson 7 – DL

3.2.9.1 Equipment

S.E. Airplane

3.2.9.2 Lesson Objective

This lesson provides a review to increase proficiency of attitude instrument flying.

3.2.9.3 Lesson Content

Preflight Procedures Instrument Flight Deck Check	Review
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb Procedures Normal Approach and Landing Procedures	 Review Review
Climbs (IR) Constant Airspeed Climbs Constant Rate Climbs	 Review Review
Flight by Reference to Instruments Straight and Level Flight Change of Airspeed Operations in Turbulence Recovery from Unusual Flight Attitudes	 Review Review Review Review
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control	 Review Review
Steep Turns (IR) Steep Turns	 Review
Turns (IR) Standard Rate Turns Timed Turns to Magnetic Compass/ Standby Instrument Headings Compass Turns to Magnetic Compass Headings (If Installed)	 Review Review Review
Descents (IR) Constant Airspeed Descents Constant Rate Descents	 Review Review
Stalls (IR) Power-Off Stalls Power-On Stalls	 Review Review
APPROACH AND LANDING Normal Approach and Landing	

3.2.9.4 Completion Standards

At the completion of this lesson, the student will have gained an increased proficiency of IFR procedures and the student will be able to immediately recognize the approach of stalls and other unusual attitudes and demonstrate correct recovery procedures. Altitude should be maintained within ±150 feet, heading within ±15oand airspeed within ±15 knots

3.2.10 Lesson 8 – DL

3.2.10.1 Equipment

S.E. AIRPLANE

3.2.10.2 Lesson Objective

This lesson provides a review to increase proficiency of attitude instrument flying.

3.2.10.3 Lesson Content

Preflight Procedures Preflight Assessment Instrument Flight Deck Check	 Review Review
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb Normal Approach and Landing	 Review Review
Climbs (IR) Constant Airspeed Climbs Constant Rate Climbs	 Review Review
Flight by Reference to Instruments Straight and Level Flight Change of Airspeed Recovery from Unusual Flight Attitudes	 Review Review Review
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control	 Review Review
Steep Turns (IR) Steep Turns	 Review
Turns (IR) Standard Rate Turns Timed Turns to Magnetic Compass/ Standby Instrument Headings Compass Turns to Magnetic Compass Headings (If Installed)	 Review Review Review
Descents (IR) Constant Airspeed Descents Constant Rate Descents	 Review Review
Stalls (IR) Power-Off Stalls Power-On Stalls	 Review Review
Postflight Procedures After Landing, Parking and Securing Checking Instruments and Equipment	 Review Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.2.10.4 Completion Standards

At the completion of this lesson, the student will have gained an increased proficiency of IFR procedures and the student will be able to immediately recognize the approach of stalls and other unusual attitudes and demonstrate correct recovery procedures. Altitude should be maintained within ±150 feet, heading within ±15oand airspeed within ±15 knots.

3.2.11 Lesson 9 – AATD/FTD

3.2.11.1 Equipment

AATD/FTD

3.2.11.2 Lesson Objective

This lesson provides an introduction to partial panel attitude instrument procedures.

3.2.11.3 Lesson Content

Preflight Procedures Instrument Flight Deck Check	 Review
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb Procedures	 Review
Climbs (IR) Constant Airspeed Climbs Constant Rate Climbs	 Review Review
Flight by Reference to Instruments Straight and Level Flight Change of Airspeed Use of Trim Use of Standby Instruments Recovery from Unusual Flight Attitudes Pattern Alpha	 Review Review Review Introduction Review Review
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control	 Review Review
Turns (IR) Standard Rate Turns Timed Turns to Magnetic Compass/ Standby Instrument Headings Compass Turns to Magnetic Compass Headings (If Installed) Climbing Turns Descending Turns	 Review Review Review Review Review
Descents (IR) Constant Airspeed Descents Constant Rate Descents	 Review Review
Stalls (IR) Power-Off Stalls Power-On Stalls	 Review Review
Instrument Approach Procedures Visual Approach Procedures	 Introduction
Additional Areas As Assigned By Instructor (If Necessary)	

3.2.11.4 Completion Standards

At the completion of this lesson, the student will maintain altitude within ±150 feet, heading within ±15o and airspeeds within ±15 knots. Using partial panel instrument reference, the student should maintain altitude within ± 200 feet, heading within ± 20 o and airspeed within ± 20 knots.

3.2.12 Lesson 10 – AATD/FTD

3.2.12.1 Equipment

AATD/FTD

3.2.12.2 Lesson Objective

This lesson provides a review of partial panel attitude instrument procedures. The flight instructor will simulate conditions requiring the student to execute all maneuvers in partial panel conditions.

3.2.12.3 Lesson Content

Preflight Procedures Instrument Flight Deck Check	 Review
Climbs (IR) Constant Airspeed Climbs Constant Rate Climbs	 Review Review
Flight by Reference to Instruments Straight and Level Flight Change of Airspeed Use of Trim Use of Standby Instruments Recovery from Unusual Flight Attitudes Pattern Bravo	 Review Review Review Review Review Review
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control	 Review Review
Turns (IR) Standard Rate Turns Constant Bank Turns Timed Turns to Magnetic Compass/ Standby Instrument Headings Compass Turns to Magnetic Compass Headings (If Installed) Climbing Turns Descending Turns	 Review Review Review Review Review
Descents (IR) Constant Airspeed Descents Constant Rate Descents	 Review Review
Stalls (IR) Power-Off Stalls Power-On Stalls	 Review Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.2.12.4 Completion Standards

At the completion of this lesson, the student will maintain altitude within ±150 feet, heading within ±15o and airspeeds within ±15 knots.

3.2.13 Lesson 11 – DL

3.2.13.1 Equipment

S.E. AIRPLANE

3.2.13.2 Lesson Objective

This lesson provides the instructor the opportunity to evaluate the student's proficiency in controlling the aircraft solely by instrument reference in both full and partial panel situations for all applicable maneuvers.

3.2.13.3 Lesson Content

Preflight Procedures Preflight Assessment Instrument Flight Deck Check	 Review Review
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb Normal Approach and Landing	 Review Review
Climbs (IR) Constant Airspeed Climbs Constant Rate Climbs	 Review Review
Flight by Reference to Instruments Straight and Level Flight Change of Airspeed Use of Standby Instruments Recovery from Unusual Flight Attitudes	 Review Review Review Review
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control	 Review Review
Turns (IR) Standard Rate Turns Timed Turns to Magnetic Compass	 Review Review
Descents (IR) Constant Airspeed Descents Constant Rate Descents	 Review Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.2.13.4 Completion Standards

At the completion of this lesson, the student will exhibit improved skills for the listed tasks. During both the full and partial panel tasks the student will be able to maintain altitude within ±150 feet, heading within ±15o, airspeeds within ±15 knots and targeted climb or descent rates within ± 200 feet per minute. During recovery from unusual attitudes any intervention by the Instructor to prevent the aircraft from exceeding any operating limitations or entering an unsafe flight condition shall be unsatisfactory.

3.2.14 Lesson 12 – GB

3.2.14.1 Equipment

Briefing Room

3.2.14.2 Lesson Objective

This lesson provides an introduction of aircraft flight instruments, navigation equipment, and operations of systems in preparation for lesson 13 and Stage I check.

3.2.14.3 Lesson Content

Very High Frequency Omni-directional Range (VOR)	
VOR components	Introduction
Function of VOR	Introduction
VOR Operational Errors	Introduction
VOR Accuracy Checks	Introduction
VOR Test Facility	Introduction
Certified Checkpoints	Introduction
Distance Measuring Equipment (DME)	
DME Components	Introduction
Function of DME	Introduction
DME Errors	Introduction
Global Positioning Systems (GPS)	
Components	Introduction
Function of GPS	Introduction
GPS Substitution	Introduction
GPS Errors	Introduction
System Status	Introduction
Wide Area Augmentation System (WAAS)	
Instrument Landing System (ILS)	
Basic Components of an ILS	Introduction
Localizer Components	Introduction
Function of Localizer	Introduction
Localizer Errors	
Glideslope/ Glide Path	
Marker Beacon & Compass Locator	
ILS Minimums	
ILS Course Distortion	
Aircraft Flight Instruments & Navigation Equipment	
Altimeter	Introduction
Airspeed Indicator	Introduction
Vertical Speed Indicator	Introduction
Attitude Indicator	Introduction
Horizontal Situation Indicator	Introduction
Heading Indicator	Introduction
Magnetic Compass (If Applicable)	Introduction
Turn & Slip Indicator/Turn Coordinator	
National Airspace System	
Controlled/Uncontrolled	Introduction
Class A, B, C, D, E, G	Introduction
Special Use Airspace	Introduction
Other Use Airspace	Introduction
Aeromedical Factors	
Human Factors	Introduction
Additional Areas	
As Assigned by Instructor (If Necessary)	

3.2.14.4 Completion Standards

At the completion of this lesson the student will have gained an increased level of knowledge in the tasks listed above.

3.2.15 Lesson 13 – AATD/FTD

3.2.15.1 Equipment

AATD/FTD

3.2.15.2 Lesson Objective

Now is the time to expand your knowledge and skills while flying in IMC. This lesson will introduce full panel VOR and GPS intercepting and tracking procedures.

3.2.15.3 Lesson Content

Preflight Procedures Instrument Flight Deck Check	 Review
Checklist Usage Normal Procedures Checklist	 Review
Flight by Reference to Instruments Pattern Bravo Recovery from Unusual Flight Attitudes	 Review Review
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control Climbs/Turns/Descents	 Review Review Review
Navigation Systems VOR Orientation VOR Intercepting & Tracking VOR Accuracy Check GPS Intercepting & Tracking Use of RMI Arc Procedures Intercepting and Navigational Systems and Arcs DME Arc Procedures	 Introduction Introduction Introduction Introduction Introduction Introduction Introduction Introduction
Additional Areas As Assigned By Instructor (If Necessary)	

3.2.15.4 Completion Standards

At the completion of this lesson, the student will have gained an increased proficiency in local instrument navigation procedures. During completion of the tasks the student will maintain altitude within ±150 feet, heading within ±15 o and airspeeds within ±15 knots.

3.2.16 Lesson 14 – AATD/FTD

3.2.16.1 Equipment

AATD/FTD

3.2.16.2 Lesson Objective

During this lesson you will continue to build on BAI by introducing full panel LOC and LOC back course intercepting and tracking.

3.2.16.3 Lesson Content

Preflight Procedures Instrument Flight Deck Check	 Review
Cockpit Familiarization Flight Controls Engine Controls	 Review Review
Checklist Usage Normal Procedures Checklist	 Review
Engine Start Engine Fire On Start Procedures	 Introduction
Flight by Reference to Instruments Recovery from Unusual Flight Attitudes	 Review
Climbs (IR) Constant Airspeed Climbs Constant Rate Climbs Climbs Using PFD	 Review Review Review
Turns (IR) Level Turns Using PFD Standard Rate Turns Compass Turns Timed Turns to Magnetic Headings	 Review Review Review Review
Navigation Systems Localizer Orientation Localizer Intercepting & Tracking Localizer/BC Orientation Localizer/BC Intercepting and Tracking VOR Accuracy Check Intercepting and Tracking Navigational Systems and Arcs	 Introduction Introduction Introduction Review Review Review
Steep Turns (IR) Steep Turns	 Review
Descents (IR) Constant Rate Descents Constant Airspeed Descents Descents Using PFD	 Review Review Review
Postflight Procedures Checking Instruments and Equipment	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.2.16.4 Completion Standards

At the completion of this lesson, the student will display increased proficiency in attitude instrument flight. During the full panel flight the student should maintain altitude within ±150 feet, heading within ±15º, and airspeed within ±15 knots. The student will exhibit a basic understanding of LOC and LOC/BC orientation and tracking procedures.

3.2.17 Lesson 15 – AATD/FTD

3.2.17.1 Equipment

AATD/FTD

3.2.17.2 Lesson Objective

This lesson provides a review of intercepting and tracking, and DME arc's and procedures both full and partial panel.

3.2.17.3 Lesson Content

Preflight Procedures Instrument Flight Deck Check	 Review
Engine Start General Starting Procedures Engine Fire On Start Procedures	 Review Review
ATC Communications ATC Communications Standard Phraseology	 Review Review
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb Procedures Normal Approach and Landing Procedures	 Review Review
Flight by Reference to Instruments Recovery from Unusual Flight Attitudes Standby Instrument Usage	 Review Review
Stalls (IR) Stalls /Recovery Procedures Autopilot Stalls (If Installed)	 Review Introduction
Navigation Systems VOR Intercepting & Tracking GPS Intercepting & Tracking Localizer/ LOC BC Orientation Localizer/ LOC BC Intercepting & Tracking Intercepting and Tracking Navigational Systems and Arcs DME Arc Procedures (If applicable)	 Review Review Review Review Review Review
Postflight Procedures Checking Instruments and Equipment	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.2.17.4 Completion Standards

At the completion of this lesson, the student will display increased proficiency in attitude instrument flight. During both full and partial panel flight the student should maintain altitude within ±150 feet, heading within ± 15 o, and airspeed within ±15 knots. Course navigation will be maintained with less than full scale deflection of the CDI. The student will exhibit a basic understanding of VOR & LOC orientation and tracking procedures, including the interception of specific VOR radials and Arcs.

3.2.18 Lesson 16 – DL

3.2.18.1 Equipment

S.E. AIRPLANE

3.2.18.2 Lesson Objective

This lesson provides a review of both partial and full panel VOR, and GPS procedures.

3.2.18.3 Lesson Content

Preflight Procedures	
Aircraft Systems Related to IFR Operations	Review
Aircraft Flight Instruments and Navigation Equipment	Review
Instrument Flight Deck Check	Review
Preflight Assessment	Review
Takeoffs, Landings, and Go-Arounds	
Normal Takeoff and Climb	Review
Normal Approach and Landing	Review
Navigation Systems	
Intercepting and Tracking Navigational Systems and Arcs	Review
VOR Orientation	Review
VOR Accuracy Check	Review
VOR Time, Speed, and Distance to a Station	Review
GPS Interception and Tracking	Review
GPS Operation	Review
Postflight Procedures	
After Landing, Parking and Securing	Review
Checking Instruments and Equipment	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

3.2.18.4 Completion Standards

At the completion of this lesson, the student will exhibit increased proficiency in all GPS and VOR tracking. The student will understand GPS & VOR orientation and be able to demonstrate while using partial panel attitude instrument procedures. The student will maintain altitude within ±150 feet, heading within ±15 o, and airspeed within ±15 knots.

3.2.19 Lesson 17 – AATD/FTD

3.2.19.1 Equipment

AATD/FTD

3.2.19.2 Lesson Objective

This lesson provides an instructor evaluation of the student's proficiency in basic attitude instrument flying and navigation using full and partial panel instrument procedures.

3.2.19.3 Lesson Content

Training Record Review of Training Folder Verify/Update Manuals Verify/Update Publications	 Review Review Review
Preflight Procedures Pilot Qualifications Performance and Limitations Instrument Flight Deck Check Taxiing Before Takeoff Check	 Review Review Review Review Review
Engine Start General Starting Procedures Engine Fire On Start Procedures	 Review Review
ATC Communications ATC Communications Standard Phraseology	 Review Review
Flight by Reference to Instruments Recovery from Unusual Flight Attitudes	 Review
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb Procedures Normal Approach and Landing Procedures	 Review Review
Turns (IR) Timed Turns to Magnetic Compass/ Standby Instrument Headings Compass Turns to Magnetic Compass Headings (If Installed)	 Review Review
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control	 Review Review
Stalls (IR) Stalls /Recovery Procedures	 Review
Intercepting & Tracking VOR Orientation VOR Intercepting & Tracking GPS Intercepting & Tracking Localizer Intercepting & Tracking	 Review Review Review
Navigation Systems Intercepting and Tracking Navigational Systems and Arcs	 Review
Steep Turns (IR) Steep Turns	 Review
Postflight Procedures Checking Instruments and Equipment	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.2.19.4 Completion Standards

At the completion of this lesson, the student will show increased proficiency for all listed tasks.

Altitude will be maintained within ±150 feet, heading within ±15º,and airspeed within ±15 knots, and targeted climb or descent rates within ±200 feet per minute. VOR, LOC, GPS and ARC will be accurate with the student maintaining position orientation at all times maintaining no more than ¾ scale deflection of the CDI using full and partial panel instrument procedures.

3.2.20 Lesson 18 – STO (Stage Check I)

3.2.20.1 Equipment

Briefing Room

3.2.20.2 Lesson Objective

During this stage check, the Chief Instructor, designated Assistant Chief or another designated Check Instructor will evaluate the student's proficiency in basic attitude instrument flying and navigation knowledge.

3.2.20.3 Lesson Content

Training Records Identity Verification Review of Training Folder Verify/Update Manuals Verify/Update Publications
Preflight Procedures Pilot Qualifications Aircraft Systems Related to IFR Operations Aircraft Flight Instruments and Navigation Equipment Instrument Flight Deck Check Preflight Assessment Human Factors

3.2.20.4 Completion Standards

At the completion of this stage check oral, the student must demonstrate adequate knowledge of the elements to each of the topics listed.

3.2.21 Lesson 19 – STF (Stage II)

3.2.21.1 Equipment

AATD/FTD

3.2.21.2 Lesson Objective

During this stage check, the Chief Instructor, designated Assistant Chief or another designated Check Instructor will evaluate the student's proficiency in basic attitude instrument flying and navigation using full and partial panel instrument procedures. It is the determination of the Instructor, designated Assistant Chief or another designated Check Instructor which procedures or maneuvers will be conducted as full or partial panel.

3.2.21.3 Lesson Content

Preflight Procedures Pilot Qualifications Performance and Limitations Instrument Flight Deck Check
Preflight Preparation Taxiing Before Takeoff Check
Cockpit Familiarization Flight Controls Engine Controls
Checklist Usage Normal Procedures Checklist
Engine Start General Starting Procedures
ATC Communications ATC Communications Standard Phraseology
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb Procedures Normal Approach and Landing Procedures
Climbs (IR) Constant Airspeed Climbs Constant Rate Climbs
Turns (IR) Standard Rate Turns Timed Turns to Magnetic Compass/ Standby Instrument Headings Compass Turns to Magnetic Compass Headings (If Installed)
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control
Stalls (IR) Stalls /Recovery Procedures
Intercepting & Tracking VOR Orientation VOR Intercepting & Tracking GPS Intercepting & Tracking Localizer Orientation Localizer Intercepting & Tracking

Flight by Reference to Instruments Recovery from Unusual Flight Attitudes
Navigation Systems Intercepting and Tracking Navigational Systems and Arcs
Steep Turns (IR) Steep Turns
Descents (IR) Constant Rate Descents Constant Airspeed Descents Descents Using PFD
Postflight Procedures Checking Instruments and Equipment
Additional Areas As Assigned By Instructor (If Necessary)

3.2.21.4 Completion Standards

At the completion of this lesson, the student will show increased proficiency for all listed tasks. Altitude will be maintained within ±150 feet, heading within ± 15 o, and airspeed within ±15 knots, and targeted climb or descent rates within ±200 feet per minute. VOR, LOC, GPS and ARC will be accurate with the student maintaining position orientation at all times maintaining no more than ¾ scale deflection of the CDI using full and partial panel instrument procedures.

3.3 Stage 2

3.3.1 Stage Objective

The objective of this stage is for the student to be introduced to instrument approach procedures, Missed Approach, and holding patterns.

If the aircraft used in this course does not have an Autopilot system installed, the student will still be required to conduct the instrument procedure "hand flown".

3.3.2 Stage Completion Standards

At the completion of this stage, the student will demonstrate to the flight instructor an increased level of proficiency in instrument approach procedures, Missed Approach, and holding patterns, as well as all IFR flight maneuvers and procedures to the proficiency level of an Instrument Rated Pilot.

> **NOTE:** In the event an **AATD/FTD** is not available, the training event prescribed can be conducted in the aircraft with the approval from the Chief Instructor/Asst. Chief or designee.

3.3.3 Lesson 20 – AATD/FTD

3.3.3.1 Equipment

AATD/FTD

3.3.3.2 Lesson Objective

This lesson provides a review of emergency and abnormal procedures through the use of simulated situations in the appropriate flight training devices for which this course is being taught. These emergency/ abnormal procedures should be given in simulated IMC and appropriately addressed.

3.3.3.3 Lesson Content

Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Engine Start	
General Starting Procedures	Review
Hot Start Procedures	Review
Emergency/Abnormal Procedures	
Systems and Equipment Malfunctions	Review
Door Open in Flight	Introduction
CAPS Deployment (If Installed)	Introduction
Electrical Trim/Autopilot Failure (If Installed)	Introduction
Pitot/Static Malfunction	Introduction
Engine Partial Power Loss	Introduction
Low Oil Pressure	Introduction
Engine Failure in Flight	Introduction
Emergency Equipment and Survival Gear	Introduction
Wing Fire in Flight	Introduction
Propeller Governor Failure (If Installed)	Introduction
Alternator Failure in Flight	Introduction
Postflight Procedures	
Checking Instruments and Equipment	Review
Additional Areas	
As Assigned by Instructor (If Necessary)	

3.3.3.4 Completion Standards

At the completion of this lesson, the student will have gained proficiency in emergency and abnormal procedures with respect for the private pilot and instrument rating practical test standards. During the Checking Instruments and Equipment, the student will be assigned specific tasks to prepare for the next flight activity.

3.3.4 Lesson 21 – GB

3.3.4.1 Equipment

Briefing Room

3.3.4.2 Lesson Objective

During this lesson the student will review Holding Procedures, Instrument Approach Procedures, Filing IFR Flight Plans and applicable publications and validity.

3.3.4.3 Lesson Content

Stage Goals & Objective Stage 2 Overview	Review
Holding Procedures Entries Speeds/Altitudes Timing/Patterns 5 T's	 Introduction Introduction Introduction Introduction
Instrument Approach Procedures Approach Briefing Precision Approaches Non-Precision Approaches Circling Approach	 Introduction Introduction Introduction Introduction
Flight Plans Filing IFR Flight Plans	 Introduction
Publications Chart Currency Terminal Charts Enroute Charts Use of Chart Supplement/ AFD Instrument Rating ACS	 Introduction Introduction Introduction Introduction Introduction
Chair Flying Instrument Approach Procedures	 Introduction
Additional Areas As Assigned by Instructor (If Necessary)	

3.3.4.4 Completion Standards

At the completion of this lesson, the student will have gained a complete knowledge of Holding Procedures, Instrument Approach Procedures, how to file an IFR flight plan and publications associated to IFR flights and the Instrument Rating ACS..

3.3.5 Lesson 22 – AATD/FTD

3.3.5.1 Equipment

AATD/FTD

3.3.5.2 Lesson Objectives

This lesson provides an introduction to the student is introduced to VOR*, GPS and/or VOR Intersection and Localizer holding patterns (standard and non-standard) (as appropriate).

*VOR holds need to be conducted at/ over a VOR Station

3.3.5.3 Lesson Content

Preflight Procedures Preflight Inspection	Review
Checklist Usage Normal Procedures Checklist	Review
ATC Communications ATC Communications Standard Phraseology	Review Review
Takeoffs, Landings, and Go-Arounds Instrument Takeoff Procedures Normal Approach and Landing Procedures	Introduction Review
Air Traffic Control Clearances and Procedures Holding Procedures-VOR* Holding Procedures-GPS Holding Procedures-VOR/Intersection Holding Procedures-LOC	Introduction Introduction Introduction Introduction
Postflight Procedures Checking Instruments and Equipment	Review
Additional Areas As Assigned by Instructor (If Necessary)	

3.3.5.4 Completion Standards

At the completion of this Lesson, the student, using the VOR, GPS, VOR/ Intersection and LOC hold, will demonstrate correct entries to holding patterns. During holding procedures, the student should be able to maintain altitude within + 150 feet, headings within + 15o, and airspeeds within + 15 knots.

3.3.6 Lesson 23 – GB

3.3.6.1 Equipment

Briefing Room

3.3.6.2 Lesson Objective

This lesson provides an introduction to IFR navigation logs, DPs, STARs, and Air Traffic Control Clearances..

3.3.6.3 Lesson Content

Training Records	
Review of Training Folder	Review
Verify/Update Manuals	Review
Verify/Update Publications	Review
Preflight Preparation	
Cross-Country Flight Planning	Introduction
Weather Information	Introduction
IFR Navigation Logs	Introduction
Preferred Routes	Introduction
Alternate Considerations	Introduction
Fuel Considerations	Introduction
Departure Procedures	
Abbreviated IFR Departure Clearance Procedures	Introduction
Hold for Release/Release Times	Introduction
Clearance Void Times	Introduction
Instrument Departure Procedures	Introduction
Standard Instrument Departure (SID)	Introduction
Obstacle Departure Procedure (ODP)	Introduction
Enroute Procedures	
Position Reports	Introduction
Mandatory Reports	Introduction
Non-Radar Reporting Points	Introduction
Additional Reports	Introduction
Airways and Route Systems	Introduction
Changeover Points	Introduction
Arrival Procedures	
STAR's	Introduction
Visual Approach Procedures	Introduction
Contact Approach	Introduction
Instrument Approach Procedures	Introduction
Approach Charts & Instrument Approach Plates	Introduction
Side-Step Maneuver	Introduction
Additional Areas	
As Assigned by Instructor (If Necessary)	

3.3.6.4 Completion Standards

At the completion of this lesson the student must have adequate knowledge of the tasks listed.

3.3.7 Lesson 24 – AATD/FTD

3.3.7.1 Equipment

AATD/FTD

3.3.7.2 Lesson Objective

This lesson provides an introduction to instrument approach procedures as well as a review of standard and non-standard holds. Proficiency is increased through the practice of instrument approach procedures.

*VOR holds need to be conducted at/over a VOR Station

3.3.7.3 Lesson Content

Checklist Usage Normal Procedures Checklist	 Review
ATC Communications ATC Communications Standard Phraseology	 Review Review
Takeoffs, Landings, and Go-Arounds Instrument Takeoff Normal Approach and Landing	 Review Review
Air Traffic Control Clearances and Procedures Holding Procedures - GPS Holding Procedures - VOR/Intersection	 Introduction Introduction
Instrument Approaches Precision Approach - ILS Nonprecision Approach - VOR* Nonprecision Approach - GPS	 Introduction Introduction Introduction
Instrument Approach Procedures Circling Approach Full Approach Missed Approach	 Introduction Introduction Introduction
Postflight Procedures Checking Instruments and Equipment	 Review
Additional Areas As Assigned by Instructor (If Necessary)	

3.3.7.4 Completion Standards

At the completion of this lesson, the student will have the necessary skill and knowledge to determine correct holding pattern entries for both standard and nonstandard holding patterns. Instrument approaches will be held to within full scale deflection of the course deviation indicator (CDI). Altitude should be maintained within + 150 feet, heading within + 15 o and airspeed within + 15 knots. While holding, one minute inbound legs + 10 seconds must be accomplished.

3.3.8 Lesson 25 – DL

3.3.8.1 Equipment

S.E. AIRPLANE

3.3.8.2 Lesson Objective

This lesson provides a review of the ILS approach, localizer approach and holding procedures

3.3.8.3 Lesson Content

Preflight Procedures Instrument Flight Deck Check	Review
Checklist Usage Normal Procedures Checklist	Review
ATC Communications ATC Communications Standard Phraseology	 Review Review
Air Traffic Control Clearances and Procedures Holding Procedures	Review
Instrument Approaches Precision Approach - ILS Nonprecision Approach - LOC	 Review Review
Instrument Approach Procedures Circling Approach Missed Approach Landing from an Instrument Approach	 Review Review Review
Postflight Procedures Checking Instruments and Equipment	Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.3.8.4 Completion Standards

At the completion of this lesson, the student will maintain altitude within +150 feet, headings within + 15o and airspeeds within + 15 knots while practicing holding procedures. Instrument approaches will be held to within full scale deflection of the course deviation indicator (CDI). During all operations other than holding procedures, the student must meet standards found in the Instrument Rating Airplane ACS for applicable tasks.

3.3.9 Lesson 26 – AATD/FTD

3.3.9.1 Equipment

AATD/FTD

3.3.9.2 Lesson Objective

This lesson provides a review of VOR, GPS, localizer, missed approach and Circling Approach.

3.3.9.3 Lesson Content

Air Traffic Control Clearances and Procedures Holding Procedures Holding Procedures - Intersection /DME Holding Procedures - LOC	 Review Review Review
Instrument Approaches Nonprecision Approach - GPS Nonprecision Approach - VOR Nonprecision Approach - LOC	 Review Review Review
Instrument Approach Procedures Circling Approach Straight-In Approach Full Approach Missed Approach	 Review Review Review Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.3.9.4 Completion Standards

The student will maintain altitude within ±150 feet, heading within ±15 , and airspeed within ±15 knots. All approaches will be held to ¾ scale deflection of the CDI and glideslope indications where appropriate. Additionally the student will exhibit increased SRM and Risk Assessment skills in all areas.

3.3.10 Lesson 27 – AATD/FTD

3.3.10.1 Equipment

AATD/FTD

3.3.10.2 Lesson Objective

This lesson provides a review of precision approaches, localizer approaches (front and back course), autopilot coupled and an introduction to partial panel approach procedures.

3.3.10.3 Lesson Content

Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb	 Review
Air Traffic Control Clearances and Procedures Compliance with Air Traffic Control Clearances Holding Procedures	 Review Review
Instrument Approaches Nonprecision Approach - Partial Panel GPS Nonprecision Approach - VOR Nonprecision Approach - LOC Nonprecision Approach - LOC/BC	 Introduction Review Review Review
Instrument Approach Procedures Circling Approach Straight-In Approach Autopilot Coupled Approach (If Installed) Full Approach Missed Approach Landing from an Instrument Approach	 Review Review Review Review Review Review
Emergency Operations Approach with Loss of Primary Flight Instrument Indicators	 Introduction
Additional Areas As Assigned by Instructor (If Necessary)	

3.3.10.4 Completion Standards

At the completion of this lesson, the student will demonstrate correct procedures for executing GPS, VOR and localizer (LOC) approaches. During the lesson, altitude will be maintained within + 150 feet, heading within + 15o and airspeed within + 15 knots. While on the final approach segment, the student will allow less than ¾ scale deflection of the CDI for both full and partial panel.

3.3.11 Lesson 28 – DL

3.3.11.1 Equipment

S.E. AIRPLANE

3.3.11.2 Lesson Objective

This lesson provides a review of precision approaches, localizer approaches , autopilot coupled and partial panel approach procedures.

3.3.11.3 Lesson Content

Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb	Review
Air Traffic Control Clearances and Procedures Holding Procedures-VOR/Intersection	Review
Instrument Approaches Precision Approach - ILS Nonprecision Approach - LOC	Review Review
Instrument Approach Procedures Autopilot Coupled (If Installed) Straight-In/Circling Approach Missed Approach	Review Review Review
Emergency Operations Approach with Loss of Primary Flight Instrument Indicators	Review
Additional Areas As Assigned by Instructor (If Necessary)	

3.3.11.4 Completion Standards

At the completion of this lesson, the student will demonstrate correct procedures for executing localizer (front and back course), and ILS approaches. During the lesson, altitude will be maintained within + 150 feet, heading within + 15o and airspeed within + 15 knots. While on the final approach segment, the student will maintain less than ¾ scale deflection of the CDI.

3.3.12 Lesson 29 – DL

3.3.12.1 Equipment

S.E. AIRPLANE

3.3.12.2 Lesson Objective

This lesson provides further review of precision approaches, localizer approaches (front and back course) and partial panel approaches.

3.3.12.3 Lesson Content

Preflight Procedures Preflight Briefing Preflight Assessment Instrument Flight Deck Check	 Review Review Review
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb	 Review
Air Traffic Control Clearances and Procedures Compliance with Air Traffic Control Clearances Holding Procedures	 Review Review
Navigation Systems Intercepting and Tracking Navigational Systems and Arcs VOR Accuracy Check	 Review Review
Instrument Approaches Partial Panel Precision Approach - ILS Nonprecision Approach - LOC	 Introduction Review
Instrument Approach Procedures Missed Approach	 Review
Approach and Landing Procedures Landing from an Instrument Approach	 Review
Postflight Procedures After Landing, Parking and Securing Checking Instruments and Equipment	 Review Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.3.12.4 Completion Standards

At the completion of this lesson, the student should demonstrate correct procedures for executing localizer (front and back course) and ILS approaches. During the lesson, altitude will be maintained within + 150 feet, heading within + 15oand airspeed within + 15 knots. While on the final approach segment, the student will allow less than ¾ scale deflection of the CDI.

3.3.13 Lesson 30 – TD/FTD

3.3.13.1 Equipment

PPP/ CRG-AATD/FTD

CAAC & CAA - S.E. AIRPLANE

3.3.13.2 Lesson Objective

This lesson provides the student with a review by using previously learned approach procedures and GPS tracking procedures to execute GPS approaches (as appropriate). In addition, partial panel skill will be further developed through practicing approaches while experiencing a partial panel condition.

3.3.13.3 Lesson Content

Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb	Review
Air Traffic Control Clearances and Procedures Holding Procedures - GPS	Review
Instrument Approaches Nonprecision Approach - GPS Partial Panel Nonprecision Approach - GPS Nonprecision Approach - VOR	 Review Review Review
Instrument Approach Procedures Circling Approach & Procedures Autopilot Coupled Full Approach & Procedures Visual Approach Missed Approach	 Review Review Review Review Review
Emergency Operations Approach with Loss of Primary Flight Instrument Indicators	Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.3.13.4 Completion Standards

At the completion of this lesson, the student will demonstrate the correct procedures used to perform GPS & VOR approaches. During the execution of all approaches, altitude will be maintained within + 150 feet, heading within + 15, airspeed within 15 knots and ¾ scale deflection.

3.3.14 Lesson 31 – DL

3.3.14.1 Equipment

S.E. AIRPLANE

3.3.14.2 Lesson Objective

This lesson provides the student opportunity to use previously learned approach procedures and GPS tracking procedures to execute GPS approaches. In addition, partial panel skill will be further developed through practicing approaches while experiencing a simulated partial panel condition..

3.3.14.3 Lesson Content

Preflight Procedures	
Aircraft Systems Related to IFR Operations	Review
Aircraft Flight Instruments and Navigation Equipment Instrument Flight Deck	Review
Check	Review
Air Traffic Control Clearances and Procedures	
Missed Approach to a Hold	Review
Takeoffs, Landings, and Go-Arounds	
Normal Takeoff and Climb	Review
Instrument Approaches	
Nonprecision Approach - GPS	Review
Precision Approach - ILS	Review
Partial Panel Nonprecision Approach - GPS	Review
Instrument Approach Procedures	
Autopilot Coupled Partial Panel Approach (If Installed)	Review
Circling Approach	Review
Full Approach	Review
Landing from an Instrument Approach	Review
Emergency Operations	
Loss of Communications	Review
Approach with Loss of Primary Flight Instrument Indicators	Review
Postflight Procedures	
After Landing, Parking and Securing	Review
Checking Instruments and Equipment	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

3.3.14.4 Completion Standards

At the completion of this lesson, the student will demonstrate the correct procedures used to perform GPS approaches and holding procedures. Altitude will be maintained within + 150 feet, heading within + 15 o and airspeed within + 15 knots. While on the final approach segment, the student will maintain no more than ¾ scale deflection of the CDI.

3.3.15 Lesson 32 – AATD/FTD

3.3.15.1 Equipment

PPP/ CRG-AATD/FTD

CAAC & CAA - S.E. AIRPLANE

3.3.15.2 Lesson Objective

This lesson will provide the student a review of full and partial panel approach procedures..

3.3.15.3 Lesson Content

Air Traffic Control Clearances and Procedures Holding Procedures-GPS	Review
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb	Review
Instrument Approaches Nonprecision Approach - GPS Partial Panel Nonprecision Approach - GPS Nonprecision Approach - VOR or LOC	 Review Review Review
Emergency Operations Approach with Loss of Primary Flight Instrument Indicators	Review
Additional Areas As Assigned By Instructor (If Necessary)	

3.3.15.4 Completion Standards

At the completion of this lesson, the student will demonstrate knowledge of the procedures used to perform non-precision and partial panel approaches. Altitude will be maintained within + 150 feet, heading within + 15 o and airspeed within + 15 knots. While on the final approach segment, the student will maintain no more than ¾ scale deflection of the CDI

3.3.16 Lesson 33 – GB

3.3.16.1 Equipment

Briefing Room

3.3.16.2 Lesson Objective

This lesson provides a review of IFR instrument approach procedures and holding in preparation for the Stage II check.

3.3.16.3 Lesson Content

Training Records	
Review of Training Folder	Review
Verify/Update Manuals	Review
Verify/Update Publications	Review
Preflight Preparation	
Pilot Qualifications	Review
Weather Information	Review
Airworthiness Requirements	Review
National Airspace System	Review
Performance and Limitations	Review
Human Factors	Review
Preflight Procedures	
Airplane Systems Related to IFR Operations	Review
Instrument Flight Deck Check	Review
Air Traffic Control Clearances and Procedures	
Compliance with Air Traffic Control Clearances	Review
Approach Charts & Instrument Approach Plates	Review
Navigation Systems	
Departure, Enroute, and Arrival Operations	Review
Emergency Operations	
Loss of Communications	Review
Additional Areas	
As Assigned by Instructor (If Necessary)	

3.3.16.4 Completion Standards

At the completion of this lesson the student will have knowledge of the tasks that meets or exceeds the desired outcome level listed above with respect to the instrument rating airplane ACS.

3.3.17 Lesson 34 – DL

3.3.17.1 Equipment

S.E. AIRPLANE

3.3.17.2 Lesson Objective

This lesson provides the student with a review by using previously learned approach procedures. In addition, partial panel skill will be further developed through practicing approaches while experiencing a partial panel condition getting the student prepared for intermediate stage check

3.3.17.3 Lesson Content

Preflight Procedures Instrument Flight Deck Check	 Review
Air Traffic Control Clearances and Procedures Compliance with Air Traffic Control Clearances Holding Procedures	 Review Review
Takeoffs, Landings, and Go-Arounds Normal Takeoff and Climb Normal Approach and Landing	 Review Review
Flight by Reference to Instruments Instrument Flight Recovery from Unusual Flight Attitudes	 Review Review
Navigation Systems Intercepting and Tracking Navigational Systems and Arcs	 Review
Instrument Approaches Precision Approach-ILS Nonprecision Approach-LOC Nonprecision Approach - GPS	 Review Review Review
Instrument Approach Procedures Circling Approach Missed Approach Landing from a Landing from an Instrument Approach	 Review Review Review
Emergency Operations Loss of Communications Approach with Loss of Primary Flight Instrument Indicators	 Review Review
Postflight Procedures After Landing, Parking and Securing Checking Instruments and Equipment	 Review Review
Additional Areas As Assigned By Instructor (If Necessary)	 Review

3.3.17.4 Completion Standards

At the completion of this lesson, the student will demonstrate the knowledge and proficiency required to progress to the final stage of the Instrument Course. Altitude will be maintained within ± 150 feet, heading within ± 15 o and airspeed within ± 15 knots. While on the final approach segment, the student will maintain no more than ¾ scale deflection of the CDI

3.3.18 Lesson 35 – STO (Stage II Check)

3.3.18.1 Equipment

Briefing Room

3.3.18.2 Lesson Objective

This lesson is an intermediate stage check conducted by the Chief Flight Instructor, designated Assistant Chief Flight Instructor or Check Instructor.

3.3.18.3 Lesson Content

Preflight Preparation Pilot Qualifications Weather Information
Preflight Procedures Airplane Systems Related to IFR Operations Airplane Flight Instruments and Navigation Equipment Instrument Flight Deck Check
Air Traffic Control Clearances and Procedures Approach Charts & Instrument Approach Plates
Emergency Operations Loss of Communications
Additional Areas As Assigned by Instructor (If Necessary)

3.3.18.4 Completion Standards

During the oral, the student must demonstrate Instrument pilot proficiency in accordance with the current FAA Instrument Rating-Airplane ACS Standards for all tasks listed..

3.3.19 Lesson 36 – STF (Stage II Check)

3.3.19.1 Equipment

S.E. AIRPLANE

3.3.19.2 Lesson Objective

This lesson is an intermediate stage check conducted by the Chief Flight Instructor, designated Assistant Chief Flight Instructor or Check Instructor.

3.3.19.3 Lesson Content

Preflight Procedures Preflight Assessment Instrument Flight Deck Check
Air Traffic Control Clearances and Procedures Compliance with Air Traffic Control Clearances Holding Procedures
Flight by Reference to Instruments Instrument Flight Recovery from Unusual Flight Attitudes
Navigation Systems Intercepting and Tracking Navigational Systems and Arcs Departure, En Route, and Arrival Operations
Instrument Approaches Nonprecision Approach - LOC Partial Panel Nonprecision Approach - GPS Precision Approach
Instrument Approach Procedures Circling Approach Full Approach Missed Approach Landing from an Instrument Approach
Emergency Operations Loss of Communications Approach with Loss of Primary Flight Instrument Indicators
Postflight Procedures Checking Instruments and Equipment After Landing, Parking and Securing
Additional Areas As Assigned By Instructor (If Necessary)

3.3.19.4 Completion Standards

During the flight, the student must demonstrate Instrument pilot proficiency in accordance with the current FAA Instrument Rating- Airplane ACS for all tasks listed..

3.4 Stage 3

3.4.1 Stage Objective

The objective of this stage is for the student to be introduced to instrument cross-country planning and procedures. In addition, the student will review previous skills in preparation for certification.

If the aircraft used in this course does not have an Autopilot system installed, the student will still be required to conduct the instrument procedure "hand flown".

3.4.2 Stage Completion Standards

At the completion of this stage, the student will demonstrate to the flight instructor an increased proficiency of all tasks. The student will continue with all instrument approaches and holding patterns, as well as all flight maneuvers and procedures to the proficiency level of an instrument rated pilot, as outlined in the current FAA Instrument Rating- Airplane ACS.

> **NOTE:** In the event an **AATD/FTD** is not available, the training event can be conducted in the aircraft with the approval from the Chief Instructor/Asst. Chief or designee.

3.4.3 Lesson 37 – GB

3.4.3.1 Equipment

Briefing Room

3.4.3.2 Lesson Objective

This lesson provides a review of IFR navigation logs, DPs, STARs, and Air Traffic Control Clearances in preparation for IFR cross country flights for this stage

3.4.3.3 Lesson Content

Preflight Preparation Cross Country Flight Planning IFR Navigation Logs Preferred Routes Alternate Considerations Fuel Considerations	 Review Review Review Review Review
Emergency Operations Loss of Communications	 Review
Additional Areas As Assigned by Instructor (If Necessary)	

3.4.3.4 Completion Standards

At the completion of this lesson the student will have adequate knowledge of the tasks listed in accordance with the Instrument Rating Airplane ACS.

3.4.4 Lesson 38 – DXC

3.4.4.1 Equipment

S.E. AIRPLANE

3.4.4.2 Lesson Objective

This lesson provides the student with a review of IFR cross-country procedures. The student will plan and conduct an IFR cross-country flight to an airport 50 NM from the airport of departure. During the flight, the student will review IFR departure, en route, and arrival procedures. This flight will be flown entirely in actual or simulated instrument conditions.

3.4.4.3 Lesson Content

Preflight Preparation Weather Information	
Cross-Country Flight Planning	Review
Performance and Limitations	Review
Navigation Log	Review
Flight Plan	Review
	Review
Preflight Procedures	
Preflight Assessment	Review
Instrument Flight Deck Check	Review
Air Traffic Control Clearances and Procedures	
Compliance with Air Traffic Control Clearances	Review
Holding Procedures	Review
	Review
IFR Navigation	
Estimated Time Enroute Calculations	Introduction
Fuel Reserve Calculations	Introduction
Weather and Alternate Considerations	Introduction
Instrument Approaches	
Nonprecision Approach-GPS Approach	Review
Precision Approach - ILS	Review
Instrument Approach Procedures	
Autopilot Coupled	Review
Missed Approach	Review
Landing from an Instrument Approach	Review
Emergency Operations	
Approach with Loss of Primary Flight Instrument Indicators	Review
Loss of Communications	Review
Inadvertent Icing Encounter	Introduction
Low Fuel Supply	Introduction
Postflight Procedures	
Checking Instruments and Equipment	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

3.4.4.4 Completion Standards

At the completion of this lesson, the student will be able to explain the departure, enroute, and arrival procedures encountered during an IFR flight. In addition, the student will know the methods used to calculate ETA's and comply with route changes that may be issued by ATC or necessitated by en route weather. The student will demonstrate a sound understanding of the procedures used in various emergency situations. All tasks above will be held to the applicable standards in the Instrument Rating ACS.

3.4.5 Lesson 39 – DXC

3.4.5.1 Equipment

S.E. AIRPLANE

3.4.5.2 Lesson Objective

This lesson provides a review of IFR cross-country operations, including departure, en route, emergency, and arrival procedures. In addition, this flight requires an **instrument approach at each airport** and involves **three (3) different kinds of approaches** with the use of navigation systems. This lesson requires a distance of at least **250 nautical miles** along airways or ATC- directed routing with one segment of the flight consisting of **at least a straight-line distance of 100 nautical miles between airports**. The cross-country flight must land at an airport greater than 50nm from the airport of departure to meet cross-country flight requirements. The flight must be conducted under actual or simulated instrument conditions.

3.4.5.3 Lesson Content

Preflight Procedures	
Preflight Briefing	Review
Instrument Flight Deck Check	Review
Air Traffic Control Clearances and Procedures	
Air Traffic Control Clearances and Procedures	Review
IFR Navigation	
Estimated Time Enroute Calculations	Review
Fuel Reserve Calculations	Review
Weather and Alternate Considerations	Review
Navigation Systems	
VOR Accuracy Check	Review
Instrument Approaches	
Precision Approach - ILS	Review
Nonprecision Approach - LOC or VOR	Review
Nonprecision Approach - GPS	Review
Instrument Approach Procedures	
Autopilot Coupled Approach	Review
Circling Approach	Review
Straight-In Approach	Review
Missed Approach	Review
Landing from an Instrument Approach	Review
Emergency Operations	
Loss of Communication	
Systems & Equipment Failures	Review
Inadvertent Icing Encounter	Review
Low Fuel Supply	Review
Postflight Procedures	
After Landing, Parking and Securing	Review
Checking Instruments and Equipment	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

3.4.5.4 Completion Standards

At the completion of this lesson, the student will have a thorough understanding of cross-country and emergency procedures and meet the requirements of 14 CFR 141 Appendix C, 4, (c). Performance will be evaluated on how well the student commands the airplane during the flight, and how accurately the student complies with ATC procedures and clearances. This lesson will be performed in accordance with the Instrument Rating Airplane ACS

3.4.6 Lesson 40 – GB

3.4.6.1 Equipment

Briefing Room

3.4.6.2 Lesson Objective

This lesson provides a review of IFR navigation logs, DPs, STARs, and Air Traffic Control Clearances in preparation for End of Course Stage Check oral..

3.4.6.3 Lesson Content

Training Records Review of Training Folder Verify/Update Manuals Verify/Update Publications Verify Endorsements	 Review Review Review Review
Instrument Pilot Knowledge Test Review Review Deficient Subject Areas	 Review
Preflight Preparation Pilot Qualifications Weather Information Cross Country Flight Planning	 Review Review
Preflight Procedures Airplane Systems Related to IFR Operations Airplane Flight Instruments and Navigation Equipment Instrument Flight Deck Check	 Review Review Review
Emergency Operations Loss of Communications	 Review
Additional Areas As Assigned by Instructor (If Necessary)	

3.4.6.4 Completion Standards

At the completion of this lesson the student will have knowledge of the tasks that meets or exceeds the desired outcome level listed above outlined in the FAA Instrument Rating Airplane ACS.

3.4.7 Lesson 41 – DL

3.4.7.1 Equipment

S.E. AIRPLANE

3.4.7.2 Lesson Objective

This lesson continues preparation for the upcoming End-of-Course Test. The student must demonstrate Instrument pilot proficiency in accordance with the current FAA Instrument Rating- Airplane ACS. (Non-Precision Approach- Must do 2 different types)

3.4.7.3 Lesson Content

Preflight Procedures	
Airplane Flight Instruments and Navigation Equipment	Review
Instrument Flight Deck Check	Review
Air Traffic Control Clearances and Procedures	
Compliance with Air Traffic Control Clearances	Review
Holding Procedures	Review
Flight by Reference to Instruments	
Instrument Flight	Review
Recovery from Unusual Flight Attitudes	Review
Navigation Systems	
Intercepting and Tracking Navigational Systems and Arcs	Review
Departure, En Route, Arrival Operations	
Instrument Approaches	
Precision Approach	Review
Non-Precision Approach	Review
Non-Precision Approach	Review
Instrument Approach Procedures	
Autopilot Coupled Approach & Procedures	Review
Circling Approach	Review
Full Approach	Review
Missed Approach	Review
Landing from an Instrument Approach	Review
Emergency Operations	
Approach with Loss of Primary Flight Instrument Indicators	Review
Loss of Communications	Review
Postflight Procedures	
After Landing, Parking and Securing	Review
Checking Instruments and Equipment	Review
Additional Areas	
As Assigned by Instructor (If Necessary)	

3.4.7.4 Completion Standards

At the completion of this lesson, the student will be able to demonstrate the knowledge and proficiency required to meet the standards outlined in the current FAA Instrument Rating Airplane ACS.

3.4.8 Lesson 42 – STO(EOC)

3.4.8.1 Equipment

Briefing Room

3.4.8.2 Lesson Objective

This lesson is the End-of-Course Test conducted by the Chief Flight Instructor, designated Assistant Chief Flight Instructor or Check Instructor. During the oral, the student must demonstrate Instrument pilot proficiency in accordance with the current FAA Instrument Rating-Airplane ACS.

3.4.8.3 Lesson Content

Training Records Review of Training Folder Verify/Update Manuals Verify/Update Publications Verify Endorsements
Instrument Pilot Knowledge Test Review Review Deficient Subject Areas
Preflight Preparation Pilot Qualifications Weather Information Cross Country Flight Planning
Preflight Procedures Airplane Systems Related to IFR Operations Airplane Flight Instruments and Navigation Equipment Instrument Flight Deck Check
Emergency Operations Loss of Communications
Additional Areas As Assigned by Instructor (If Necessary)

3.4.8.4 Completion Standards

At the completion of this evaluation, the student will be able to demonstrate the knowledge and proficiency required to meet the standards outlined in the current FAA Instrument Rating Airplane ACS

3.4.9 Lesson 43 – STF(EOC)

3.4.9.1 Equipment

S.E. AIRPLANE

3.4.9.2 Lesson Objective

This lesson is the End-of-Course Test conducted by the Chief Flight Instructor, designated Assistant Chief Flight Instructor or Check Instructor. During the flight, the student must demonstrate Instrument pilot proficiency in accordance with the current FAA Instrument Rating-Airplane ACS. (Non-Precision Approach-Must do 2 different types)

3.4.9.3 Lesson Content

Preflight Procedures Airplane Flight Instruments and Navigation Equipment Instrument Flight Deck Check
Air Traffic Control Clearances and Procedures Compliance with Air Traffic Control Clearances Holding Procedures
Flight by Reference to Instruments Instrument Flight Recovery from Unusual Flight Attitudes
Navigation Systems Intercepting and Tracking Navigational Systems and Arcs (If published available) Departure, En Route, Arrival Operations
Instrument Approaches Precision Approach Non-Precision Approach Non-Precision Approach
Instrument Approach Procedures Autopilot Coupled Approach & Procedures Circling Approach Full Approach Missed Approach Landing from an Instrument Approach
Emergency Operations Approach with Loss of Primary Flight Instrument Indicators Loss of Communications
Postflight Procedures After Landing, Parking and Securing Checking Instruments and Equipment
Additional Areas As Assigned By Instructor (If Necessary)

3.4.9.4 Completion Standards

At the completion of this evaluation, the student will be able to demonstrate the knowledge and proficiency required to meet the standards outlined in the current FAA Instrument Rating Airplane ACS.

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